

Sperm fertility in mice with oligo-astheno-teratozoospermia restored by *in vivo* injection and electroporation of naked mRNA

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eLife Assessment

This study reports an approach for restoring sperm motility in mice. The strength lies in the novelty of the methodology being developed, but the evidence for the success of the method or its mechanism is **inadequate**. Additional experimental support would be required to support the conclusions of the authors.

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Abstract

Oligo-astheno-teratozoospermia (OAT), a recurrent cause of male infertility, is the most frequent disorder of spermatogenesis with a probable genetic cause. Patients and mice bearing mutations in the *ARMC2* gene have a decreased sperm concentration, and individual sperm show multiple morphological defects and a lack of motility – a canonical OAT phenotype. Intracytoplasmic sperm injection (ICSI) is required to treat such a condition but it is associated with a small increase in birth defects in comparison to pregnancies not involving assisted conception. Consequently, new targeted treatments are needed to restore fertility. Here, a combination of *in vivo* injection and electroporation of capped and poly-A-tailed naked mRNA is tested as a strategy to treat *ARMC2*-related infertility in mouse. mRNAs coding for several reporter genes are tested and the efficiency and the kinetic of expression

are assessed using *in vivo* and *in vitro* 2D and 3D imaging experiments. We show that mRNA-coded reporter proteins are detected for up to 3 weeks in germ cells, making the use of mRNA possible to treat infertility. We compare these results with those obtained with a non-integrative plasmid Enhanced Episomal Vector (EEV), which induces low and transient expression in spermatogenic cells. Consequently, injection and electroporation of naked mRNA-*Armc2* into the testes of *Armc2*-deficient males were performed and we show the presence of normal and motile sperm in the epididymis. These motile sperm were able to produce embryos by IVF and ICSI. This study shows for the first time that mRNA-*Armc2* efficiently restores fertility and opens new paths for male infertility treatment.

Introduction

Worldwide, 10-15 % of couples (or 70 million) face infertility [1]. Infertility is thus a major public health issue presenting significant medical, scientific and economic challenges (a multibillion € annual market)[2]. A significant proportion of infertilities is due to altered gametogenesis, where the sperm and eggs produced are incompatible with fertilization and/or embryonic development [3]. Approximately 40 % of cases of infertilities involve a male factor, either exclusively, or associated with a female deficiency [4].

Male gametogenesis, or spermatogenesis, is a highly complex physiological process which can be split into three successive steps: proliferation (mitosis of spermatogonia), reduction of the number of chromosomes (meiosis of spermatocytes), and morphogenesis of spermatozoa (spermiogenesis). The whole process involves around two thousands of genes, 60 % of which are expressed exclusively in the testes [5]. Because of this multiplicity of genes, spermatogenesis is strongly affected by genetic factors [5], with most severe disorders likely to be of genetic origin.

Among infertility disorders, oligo-astheno-teratozoospermia (OAT) is the most frequent (50 % [6]) and it is likely to be of genetic origin. Spermatocytograms of OAT patients show a decrease in sperm concentration, multiple morphological defects and defective motility [7, 8]. Because of these combined defects, patients are infertile and can only conceive by intracytoplasmic sperm injection (ICSI).

ICSI can efficiently overcome the problems faced. Nevertheless, concerns persist regarding the potential risks associated with this technique, including blastogenesis defect, cardiovascular defect, gastrointestinal defect, musculoskeletal defect, orofacial defect, leukemia, central nervous system tumors, and solid tumors [9–12]. Statistical analyses of birth records have demonstrated an elevated risk of birth defects, with a 30-40 % increased likelihood in cases involving ICSI [9–12], and a prevalence of birth defects between 1 % and 4 % [11]. To overcome these drawbacks, a number of experimental strategies have been proposed to bypass ARTs and restore spermatogenesis and fertility, including gene therapy [13–16].

Gene therapy consists of introducing a DNA sequence into the genome to compensate for a defective gene. It can thus rescue production of a missing protein, and is now applied both in research [17] and for the treatment of human diseases [18].

Given the genetic basis of male infertility, the first strategy, tested in mice, was to overcome spermatogenic failure associated with monogenic diseases by delivery of an intact gene to deficient germ cells [13]. Gene therapy is effective in germ cells, as numerous publications have shown that conventional plasmids can be transferred into spermatogonia in several species with success, allowing their transcription in all cells of the germinal lineage [13–16]. Most experiments were performed in mouse models, delivering DNA constructs into living mouse germ cells by testis electroporation after microinjection of a DNA-containing solution into the seminiferous tubules. Using this method, it was possible to rescue meiosis and fertility in mouse

models of infertility [13, 16]. However, the genetic changes induced are transmitted to any descendants. Consequently, gene therapy cannot be used to treat infertility in humans, both for ethical reasons and to avoid any eugenic deviations, and currently transmissible changes in humans are illegal in 39 countries [19]. Furthermore, the genetic modification of germ cell lines poses biological risks, including the induction of cancer, off-target effects, and cell mosaicism. Errors in editing may have adverse effects on future generations. It is exceedingly challenging to anticipate the consequences of genetic mosaicism, for instance, in a single individual [20, 21]. Gene therapies have thus raised both ethical controversy and long-term safety issues.

For these reasons, we decided to test an alternative strategy to DNA transfection based on the use of naked mRNA. Thanks to this change, the risk of genomic insertion is avoided, and thus there is no question of heritable alterations [22]. The first part of this study presents a characterization of the protein expression patterns obtained following transfection of naked mRNA coding for reporter genes into the testes of mice. The second part is to apply the protocol to a preclinical mouse model of OAT. Patients and mice carrying mutations in the *ARMC2* gene present a canonical OAT phenotype and are infertile. The preclinical *Armc2* deficient (*Armc2* KO) mouse model is therefore a valuable model to assess whether *in vivo* injection of naked mRNA combined with electroporation can restore spermatogenesis. We chose this model for several reasons: first, *Armc2* KO mice are sterile and all sperm exhibit short, thick or coiled flagella [13]. As a result, 100 % of sperm are immobile, thus it should be easy to determine the efficacy of the technique by measuring sperm motility with a CASA system. Second, the *Armc2* gene codes for an 867-amino acid protein and this large size represents a challenge for expression in the testis following electroporation.

To determine the efficacy of naked mRNA transfection as a method to achieve protein expression in the testes, we first assessed the level of transcription of reporter proteins after mRNA injection compared to the injection of a non-integrating plasmid, the Enhanced Episomal Vector (EEV). EEV is a vector derived from Epstein-Barr virus, it includes an origin of replication (EBV Ori) and the EBNA1 protein. Both elements, allows the synchronous initiation of extra-chromosomal EEV replication with host DNA at each S phase of the cell cycle and the segregation of the EEV episome in daughter cells. It is notable that EEV is maintained at a rate of 90-95% per cell division. It does not integrate or modify the host genome [11, 12].

In the present *in vivo* work, we injected and electroporated three distinct mRNAs coding for the following reporter proteins: green fluorescent protein (GFP), luciferase (Luc) and mCherry, and an EEV episome vector containing the sequences coding for both GFP and luciferase reporter proteins. The initial step was to characterize and validate the method of injection in adult males. Subsequently, the kinetics and patterns of expression of the electroporated mRNAs and EEV were compared using a variety of methods, including whole testis imaging, *in vivo* bioluminescence imaging, and tissue clearing. Subsequently, the mRNA transfection protocol was tested in a preclinical mouse model of OAT with the objective of restoring fertility.

Materials and methods

Animals

All procedures involving animals were performed in line with the French guidelines for the use of live animals in scientific investigations. The study protocol was approved by the local ethics committee (ComEth Grenoble # 318) and received governmental authorization (ministerial agreement # 38109-2022072716142778).

The animals used were (a) B6D2 F1 hybrid (♀ C57BL/6JRj crossed with ♂ DBA/2, Janvier laboratories; France) adult male mice aged between 8 and 25 weeks, (b) the *Armc2* KO mouse strain obtained by CRISPR-Cas9 [23] and (c) CD1 female 6 weeks old (Janvier laboratories, Le Genest-Saint-Isle, France). Experiments were carried out on wild type (WT) or *Armc2* KO adult male mice aged between 8 and 15 weeks.

Chemicals and reagents

Fast Green (FG) (F7258 – 25 g), phosphate buffered saline (PBS, D853 7-500 mL), hematoxylin (GH5316 – 500 mL), eosin (HT110216 – 500 mL), tert-Butanol (471712 – 100 mL), Histodenz (D2158 – 100 g), sorbitol (S1876 – 100 g), urea (U5128 – 500 g), Potassium Phosphate, Monobasic (P0662), Magnesium Sulfate (anhydrous) (M9397), Sodium Bicarbonate Calcium Chloride · 2H₂O (22317), EDTA (E9884), Sodium Lactate (60 % syrup - d = 1,32 g L⁻¹) (L7900), Glucose (G8270), Penicillin (P4333), Streptomycin (P4333), HEPES (H0887), PVA 30,000-70,000 (P8136), Albumin, Bovine Fraction V (A3803), M2 medium (M7167), Hyaluronidase (H3884), mineral oil (M8410), PolyVinylPyrolidone (PVP360-100G), M16 medium (MR-016) and Collagenase (C8176) were purchased from Sigma Aldrich (Saint-Quentin-Fallavier, France). Sodium Chloride (1112-A) was purchased from Euromedex (Souffelweyersheim, France). Potassium Chloride (26764) L-Glutamine (35050038), Sodium Pyruvate (11360039), NEAA (11140050) and EAA (11130036) were purchased from Life Technologies, (Waltham, MA USA). Schorr staining solution was obtained from Merck (Darmstadt, Germany). MfeI HF (R35895) and RNase-free DNase I (M03035) were obtained from New England Biolabs (Ipswich, MA, USA). Paraformaldehyde (PFA, 15710) was obtained from Electron Microscopy Science (Hatfield, PA, USA). Ketamine and xylazine were obtained from Centravet (Dinan, France). Fluorescent i-particles (NIRFiP-180) were obtained from Adjuvatis (Lyon, France). Script Gurad RNase CleanCap AG (040N-7113-1), CleanCap *EGFP*-mRNA (040L-7601-100), T7-FlashScribe kit (C-ASF3507), poly(A) tail (C-PAP5104H) and CleanCap *Luciferase*-mRNA (L-7602-1000) were obtained from Tebubio (Le Perray en Yvelines, France).

Plasmids

All plasmids, *EEV CAGs-GFP-T2A-Luciferase*, (EEV604A-2; System Bioscience, Palo Alto, CA, USA), mCherry plasmid (given by Dr. Conti MD at UCSF, San Francisco, CA, USA) and *EEV-Armc2-GFP* plasmid (CUSTOM-S017188-R2-3, Trilink, San Diego, CA, USA) were amplified by bacterial transformation (*E. coli*, EC0112; Thermo Fisher, Courtaboeuf, France). After expansion, plasmids were purified with a NucleoBond Xtra Midi kit (740410-50; Macherey-Nagel, Düren, Germany) using manufacturer's protocol. All plasmids DNA pellets were solubilized in (DNase- and RNase-free) milliQ water (Sigma Aldrich, Saint-Quentin-Fallavier, France), before being used.

The *EEV CAGs-GFP-T2A-Luciferase* episome contains the cDNA sequences of Green Fluorescent Protein (GFP) and luciferase, under the control of a CAGs promoter (Supp Fig 1). After purification, the *EEV CAGs-GFP-T2A-Luciferase* plasmid concentration was adjusted to 9 µg µL⁻¹. Prior to injection, 3.3 µL of this plasmid solution was mixed with 1 µL 0.5 % Fast Green and 5.7 µL sterile PBS to obtain a final *EEV* concentration of 3 µg µL⁻¹. The *EEV-Armc2-GFP* plasmid contains the mouse cDNA sequences of *Armc2* (*ENSMUST00000095729.11*) and the Green Fluorescent Protein (GFP) genes under the control of a strong CAGs promoter (Supp Fig 1). After amplification and purification, the final plasmid concentration was adjusted to 9 µg µL⁻¹ in water. Prior to injection, 3.3 µL of this plasmid solution was mixed with 1 µL of 0.5 % Fast Green and 5.7 µL of sterile PBS to obtain a final *EEV* concentration of 3 µg µL⁻¹. The *mCherry* plasmid contains the cDNA sequence of *mCherry* under the control of CMV and T7 promoters (Supp Fig 1). After amplification and purification, the final plasmid concentration was adjusted to 9 µg µL⁻¹.

Armc2-mRNA

Armc2-mRNA used for *in vivo* testes microinjection and electroporation was obtained from Trilink (San Diego, CA, USA). The commercial *Armc2*-mRNA has an AG CleanCap, a poly (A) tail of 120 adenosines and 3 HA-FLAG. The main challenge with mRNA-based therapy is mRNA stability. To improve mRNA stability *in vivo* and avoid its degradation by ribonucleases, optimization techniques were implemented. Thus, the used mRNA has codon optimization, a poly (A) tail and a CleanCap. To verify the efficiency of cell transfection, an *EGFP*-mRNA was injected together with the *Armc2*-mRNA. During the injection, the concentration of *EGFP*-mRNA and *Armc2*-mRNA were 300 ng μL^{-1} each.

mCherry-mRNA transcription *in vitro*

The circular *mCherry* plasmid was linearized using the restriction enzyme MfeI HF at 37 °C for 1 h. The pm-*mCherry* was then extracted and purified with the DNA Clean and Concentrator- 25 TM kit (D4033; Zymo Research, Irvine, CA, USA). The pm-*mCherry* was subsequently transcribed *in vitro* using the T7-FlashScribe kit (C-ASF3507; Tebubio, Le Perray en Yvelines, France). The mRNA was capped with a clean cap (CleanCap AG; 040N-7113-1, Tebubio, Le Perray en Yvelines, France), and a poly(A) tail (C-PAP5104H; Tebubio, Le Perray en Yvelines, France) was added before purification using the NucleoSpin RNA Clean Up kit (740948-50; Macherey-Nagel, Düren, Germany). Prior to injection, *mCherry*-mRNA was mixed with Fast Green to obtain a final concentration of 300 ng μL^{-1} (0.05 % FG, PBS).

Agarose gel electrophoresis of the Episomal Vector EEV and mRNA-mCherry

Before loading on a pre-stained (Gel Green) 1.5 % agarose gel, the EEV-plasmid and mRNA were mixed with a loading dye (Promega Blue Orange G1904, Promega, Charbonnières France). An aliquot of each sample (500 ng) was loaded into each well and electrophoresis was performed for 30 min at 100 V at room temperature (RT). A DNA size marker (Gene ruler SM1331, Thermo Scientific, Courtaboeuf, France) was used to assess molecular weight. Gel images were acquired using a ChemiDoc XRS+ (BIORAD, Marnes-la-Coquette, France).

In vivo microinjection and electroporation of testes

Electroporation was conducted as previously described [15]. Briefly, male mice B6D2, *Armc2*^{+/+} or *Armc2*^{-/-}, depending on the experimental conditions, were anesthetized with ketamine/xylazine solution (100 mg μL^{-1} and 10 mg μL^{-1} respectively). The testes were pulled out of the abdominal cavity or scrotum. Under a binocular microscope and using microcapillaries pipettes (FemtoTip II ®, Eppendorf, Montesson, France), 10 μL DNA (3 μg μL^{-1} - 0.05 % FG) or mRNA (300 ng μL^{-1} - 0.05 % FG) was injected into the seminiferous tubules through the *rete testis* applying a constant pressure (microinjector, Femto Jet 4i, Eppendorf, Montesson, France). Two series of 8 square electric pulses (25 V for 50 ms) were applied to the testis using tweezer-type electrodes linked to an electroporator (Gemini, BTX, Holliston, MA, USA). The testes were then replaced in the abdominal cavity, and the abdominal wall and skin were sutured. For each animal, the left testis was injected and electroporated with the different nucleic acids (mRNA, EEV), whereas the right testis was injected with a control solution (PBS, 0.5 % FG) as a control. Both testes were electroporated.

Flash Freezing and Fluorescence Analysis of testes

Depending on the experimental condition, 1-, 3- or 7-days post injection, the testes were collected and washed for 5 min in PBS. Then, they were embedded in Peel-A-Way Cryomolds filled with OCT Mounting media (VWR, Rosny-sous-Bois, France). The samples were flash frozen in a 100 % isopentane solution (524391; Carlo ERBA, Val-de-Reuil, France), pre-cooled with liquid nitrogen. Once frozen, they were cut into 20 μm sections using a cryostat. The cryostat sections were then

fixed with 4 % PFA-PBS for 10 min at 4 °C and counterstained with 1.8 μ M DAPI (nuclear stain) before observation using an Axioscan Z1 slide scanner or a confocal microscope LSM710 (NLO – LIVE7 – Confocor 3). The fluorescence of the different reporter proteins was detected using appropriate filters for DAPI, GFP, Texas Red (for mCherry), and Cy7 (for the Fluorescent i-particles NIRFiP-180).

Tissue collection and histological analysis

For histological analysis, treated and control B6D2 testes were fixed by immersion in 4 % paraformaldehyde (PFA) for 14 h. They were then embedded in paraffin before cutting into 5 μ m sections using a microtome (Leica biosystems, Wetzlar, Germany). After deparaffination, the sections were stained with hematoxylin and eosin. Stained sections were digitized at 20x magnification using an axioscan Z1-slide scanner (Zeiss, Jna, Germany). Spermatogenesis was assessed by measuring the area of seminiferous tubules and the cross sections of round tubules (μ m²) ($n > 35$ tubules per testis section; $n=5$ testis sections per condition). Statistical significance of differences was determined using a Student's *t*-test.

Ex vivo Fluorescence Analysis

To analyze the expression of the reporter proteins GFP and mCherry in seminiferous tubules, whole testes were examined under an inverted microscope (CKX53, Olympus, Shinjuku, Tokyo, Japan). Exogenous fluorescence was detected using filters for GFP and Texas Red.

Harris-Shorr sperm Analysis

Sperm were collected from the caudae epididymides of mice (Control, injected with EEV-GFP, GFP-mRNA, or *Armc2*-mRNA). They were allowed to swim for 10 min at 37 °C in 1 mL M2 media. Sperm were centrifuged at 500 g, washed once with PBS, and fixed 4 % paraformaldehyde in PBS for 1 min at RT. After washing with 1 mL acetate ammonia (100 mM), the sperm suspensions were spotted onto 0.1 % poly L-lysine precoated slides (Thermo Scientific, Courtaboeuf, France) and left to dry. Harris-Schorr staining was then performed according to the WHO protocol [24], and at least 150 sperm were analyzed per animal.

Whole Testis Optical clearing and 3D image reconstructions

Optical clearing (adapted from uDISCO and Fast 3D clear protocols)

Adult mice were euthanized by cervical dislocation and then transcardiac perfused with 1X PBS (Sigma Aldrich, Saint-Quentin-Fallavier, France). The testes were extracted and fixed for two days at 4 °C in 4 % PFA (Electron Microscopy Sciences, Hatfield, PA, USA). Samples were then washed with PBS for at least two days. Mouse testes were subsequently dehydrated in graded series of tert-butanol solutions (Sigma Aldrich, Saint-Quentin-Fallavier, France) at 35 °C as follows: 30 % overnight (O/N), 50 % for 24 h, 70 % for 10 h, 80 % O/N, 90 % for 10 h, 96 % O/N, and 100 % for 10 h. The testes were cleared in clearing solution (96 % Histodenz, 2 % Sorbitol, 20 % Urea) for 2 days. Then, nuclei were stained with 3.6 μ M DAPI (20 % DMSO; 2 % Triton, 1 % BSA) for 2 days. Finally, the testes were then conserved in the clearing solution at 4°C until observation by microscopy. All these steps were carried out under agitation and protected from light.

3D tissue Imaging

The optically-cleared mouse testes were imaged on a lightsheet fluorescence microscope (Blaze, Miltenyi Biotec, Germany), using a 4x NA 0.35 MI PLAN objective protected by a dipping cap for organic solvents, with an overall working distance of 15 mm. Acquisitions on the horizontal plane

were obtained with a fixed lightsheet thickness of 3.9 μm at both 488 nm and 561 nm with no overlap between horizontal planes. Voxel resolution $x = 1.21$; $y = 1.21$; $z = 2 \mu\text{m}$. 3D reconstructions were created using Imaris software (Oxford Instruments plc, Abingdon, UK).

Cellular image analysis

The optically-cleared mouse testes were imaged using a ‘ConfoBright’ system which is a unique adaptive optics confocal microscope (Nikon A1R MP, Nikon Europe B.V., The Netherlands) equipped with a deformable mirror module (AOS-micro, AlpaO, Montbonnot, France) to correct geometrical aberrations. Indeed, the 3D confocal imaging of cleared sample requires long distance objectives for deep tissue imaging, resulting in optical aberrations due to inhomogeneous refractive index. The ‘ConfoBright’ microscope corrects these geometric optical aberrations both in excitation and detection light path and restores locally the high diffraction-limited spatial resolution and the best possible photon collection efficiency. The driving metrics based on molecular brightness was used to optimize the adaptive optics in the isoplanatic patch of ca. 100 μm at each imaged depth. Images were acquired using an apo LWD 40x/1.15 water immersion objective (WD 600 μm) and a Plan apo 10x/0.45 objective. FIJI software (open-source software[25]) was used to process and analyze images and Imaris software for the 3D reconstructions.

Bioluminescence imaging

In vivo Bioluminescence imaging was performed at several time points after *in vivo* Luciferase - mRNA or EEV-GFP-luciferase injection and electroporation ($n=5$ mice per condition). Ten minutes before imaging, mice received an intraperitoneal injection of 150 $\mu\text{g g}^{-1}$ of D-luciferin (Promega, Charbonnières France), and were then anesthetized (isoflurane 4 % for induction and 1.5 % for maintenance) before placing in the optical imaging system (IVIS Lumina III, PerkinElmer, Courtaboeuf, France). *In vivo* bioluminescence signals were measured in selected regions of interest (injected testes) and were expressed as mean photons per second (mean $\pm$ SEM). Background bioluminescence was measured on images from control mice. When the *in vivo* bioluminescence signal was no longer detectable, testes were collected and immersed in a luciferin solution for a few minutes before performing *ex vivo* imaging to confirm the absence of signal.

Sperm motility

The cauda epididymis was dilacerated in 1 mL of M2 medium (Sigma Aldrich, Saint-Quentin-Fallavier, France) and spermatozoa were allowed to swim out for 10 min at 37 °C. After incubation, 30 μL of the sperm suspension was immediately placed onto analysis chamber (2X- CEL Slides, 80 μm depth, Leja Products B.V., The Netherlands) kept to 37°C for microscopic quantitative study of sperm movement. Motility of the spermatozoa was evaluated at 37°C with an Olympus microscope and Computer Aided Sperm Analysis (CASA) (CEROS II apparatus; Hamilton Thorne, Beverley, MA, USA). The settings employed for analysis were: acquisition rate: 60 Hz; number of frames: 30; minimum contrast: 30; minimum cell size: 4; low-size gate: 0.13; high-size gate: 2.43; low-intensity gate: 0.10; high-intensity gate: 1.52; minimum elongation gate: 5; maximum elongation gate: 100; magnification factor: 0.81.

The motility parameters measured were: straight line velocity (VSL); curvilinear velocity (VCL); averaged path velocity (VAP); amplitude of lateral head displacement (ALH); beat cross frequency (BCF); linearity (LIN); straightness (STR). Hyperactivated sperm were characterized by $VCL > 250 \mu\text{m s}^{-1}$, $VSL > 30 \mu\text{m s}^{-1}$, $ALH > 10 \mu\text{m}$ and $LIN < 60$, Intermediate by $VCL > 120 \mu\text{m s}^{-1}$ and $ALH > 10 \mu\text{m}$, progressive sperm by $VAP > 50 \mu\text{m s}^{-1}$ and $STR > 70 \%$ and slow sperm by $VAP < 50 \mu\text{m s}^{-1}$ and $VSL < 25 \mu\text{m s}^{-1}$.

Western blot

Western blotting was performed on HEK-293T cells (ATCC, Manassas, VA, USA) transfected with *EEV-Armc2* or *Arm2*-mRNA. Cells were transfected using JetPrime (101000027; Polyplus Illkirch, France) for DNA and JetMessenger (101000056; Polyplus Illkirch, France) for mRNA vectors, both according to the supplier's recommendations. After 48 h, the cells were washed with PBS and scraped off before centrifuging at 4 °C, 1500 RPM for 5 min. The cell pellet was then resuspended in lysis buffer (87787; Thermo Scientific, Courtaboeuf, France) supplemented with an EDTA-free cocktail of protease inhibitors (11836170001; Roche, Bale, Swiss). The suspension was stirred at 4 °C for 2 h, and then centrifuged at 16,000 g at 4 °C for 10 min. The protein content of the supernatant was estimated with QuantiPro™ BCA Assay kit (Sigma Aldrich, Saint-Quentin-Fallavier, France) before adding 5X Laemmli + 5 % β- mercaptoethanol and heating at 95 °C for 10 min. For the Western blot, 30 µg of proteins were deposited on a ready-made Bis-Tris gel 12 % (Thermo Fisher, Courtaboeuf, France). After the transfer, the PVDF membrane was blocked with 5 % milk in Tris-Buffered Saline solution containing 0.1 % Tween 20 (TTBS) before immunodetection. The anti HA antibody (11867423001; Sigma Aldrich, Saint-Quentin-Fallavier, France) was diluted in TTBS at 1/5000. After incubation with secondary antibodies (AP136P; Sigma Aldrich, Saint-Quentin-Fallavier, France) diluted at 1:10,000 in TTBS, binding was revealed with an enhanced chemiluminescence detection kit (1705062; Clarity Max Western ECL Substrate; BIORAD, Marnes-la-Coquette, France). Membranes were imaged on a ChemiDoc™ system (BIORAD, Marnes-la-Coquette, France).

Immunofluorescence

Immunofluorescence analysis of dissociated testicular cells was performed as follows. After collection, the tunica albuginea was removed from the testes. Then the tissue was mechanically separated with 18G needles. Once washed with PBS, the testicular cells were placed in a dissociation medium containing 1 mg collagenase type V /mL RPMI for 20 min at 37 °C. After filtration (100 µm filter) and centrifugation (5 min at 200 g) the pellets were resuspended in PBS before centrifugation once again. The pellet was then fixed in 1 mL PFA 4 % for 5 min. Then 10 mL of ammonium acetate (0.1 M) was added. Finally, 2 mL of medium was spread on a slide. Testicular cells were permeabilized with 0.1 % PBS-Triton X-100 for 20 min at room temperature. Slides were then blocked in 10 % BSA with PBS-Tween 0.1 % for 30 min before incubating overnight at 4 °C with primary antibodies anti-rabbit *ARMC2* (1/50) (HPA053696, Sigma Aldrich, Saint-Quentin-Fallavier, France) and anti-guinea pig tubulin (1/100) (AA345, Swiss Institute of Bioinformatics, Lausanne, Swiss) diluted in PBS-Tween 0.1 % - 5 % BSA. Slides were washed with PBS before incubating for 1 h with secondary antibodies: anti-guinea pig (1/500) (A-11073, Thermo Fischer, Courtaboeuf, France) and anti-rabbit (1:1000) (A-11036, Thermo Fischer, Courtaboeuf, France). Samples were counterstained with DAPI and mounted with DAKO mounting media (NC2313308; Life Technology, Courtaboeuf, France). Fluorescence images were acquired under a confocal microscope (Zeiss, Jena, Germany) fitted with a 63× oil immersion objective. Images were analyzed with ZEN lite software (Zeiss, Jena, Germany).

Intra-cytoplasmic sperm injection (ICSI)

Collection of gametes for ICSI

Arm2^{-/-} sperm or *Arm2*^{-/-} rescued sperm were harvested by dilaceration of the cauda epididymis. They were allowed to swim for 10 min at 37°C in 1 ml of CZB. HEPES medium containing (in mM) (HEPES 20, NaCl 81.6, KCl 4.8, MgSO₄ 1.2, CaCl₂ 1.7, KH₂PO₄ 1.2, EDTA.Na₂ 0.1, Na-lactate 31, NaHCO₃ 5, Na-pyruvate 0.3, polyvinyl alcohol 0.1 mg mL⁻¹, phenol red 10 mg mL⁻¹ (0.5 % (w/v) in DPBS), pH 7.4) KCl 125, NaCl 2.6, Na₂HPO₄ 7.8, KH₂PO₄ 1.4 and EDTA 3 (pH 7.0)). The sperm head was separated from the tail by applying multiple piezo pulses (PiezoXpert®, Eppendorf, Montesson, France)

Eggs preparation

CD1 female mice, 7 weeks old, were superovulated by IP injection of 7.5 IU pregnant mare's serum gonadotrophin (PMSG; Centravet, Dinan, France) followed by 7.5 IU HCG (Centravet, Dinan, France) 48 h later. Eggs were collected from oviducts about 14 h after HCG injection. Cumulus cells were removed with 0.1 % hyaluronidase in M2 medium for 5–10 min. Eggs were rinsed thoroughly and kept in KSOM medium containing (in g L⁻¹: NaCl 5.55, KCl 0.19, KH₂PO₄ 0.05, MgSO₄ 0.05, NaHCO₃ 2.1, CaCl₂ · 2H₂O 0.250, EDTA 0.004, L-Glutamine 0.146, Na-lactate 1.870, Na-pyruvate 0.020, Glucose 0.04, Penicillin 0.05, Streptomycin 0.07, BSA 1.000, NEAA 0.5 % and EAA 1 %.), at 15°C for at least 15 min until required for ICSI.

ICSI procedures

ICSI was performed according to the method described by Yoshida and Perry (2007)[26]. For microinjection, *Armc2*^{-/-} sperm or *Armc2*^{-/-}-rescued motile sperm heads were separated from the tail by applying multiple piezo pulses (PiezoXpert®, Eppendorf, Montesson, France). Sperm heads were introduced into the ooplasm using micromanipulators (Micromanipulator InjectMan®, Eppendorf, Montesson, France) mounted on an inverted Nikon TMD microscope (Nikon, Minato-ku, Tokyo, Japan). Eggs that survived the ICSI procedure were incubated in KSOM medium at 37 °C under in an atmosphere of 5 % CO₂. Pronucleus formation was checked at 6 h after ICSI, and outcomes were scored up to the blastocyst stage.

In vitro fertilization (IVF)

Armc2^{-/-} sperm or *Armc2*^{-/-} rescued sperm were harvested by dilaceration of the cauda epididymis. They were allowed to swim in IVF well in capacitated media (M16 + 4 % BSA) at 37 °C for 10 minutes. Eggs were collected from mature CD1 females (6 weeks old) synchronized with 7.5 units of pregnant mare serum gonadotrophin (PMSG) and 7.5 units of human chorionic gonadotrophin (hCG) prior to collection. Cumulus were incubated for 10 minutes in 500 µL M16 (MR-016; Sigma Aldrich, Saint-Quentin-Fallavier, France) / 1mg L⁻¹ collagenase (C8176, Sigma Aldrich). Cumulus-free and zona-free eggs were collected, and rinsed twice with 500 µl M16 medium. Eggs were incubated with either *Armc2*^{-/-} sperm or *Armc2*^{-/-} rescued sperm) in capacitated medium (37 °C, 5 % CO₂) for 4 hours. After incubation, unbound sperm were washed away and eggs were incubated with KSOM at 37 °C, 5 % CO₂. Twenty-four hours after fertilization, we scored the different stages (unfertilized oocytes, aborted embryos, and 2-cell embryos as an indication of successful fertilization).

Statistical analyses

Statistical analyses were performed using SigmaPlot (version 10; Systat Software, Inc., San Jose, CA, USA). To account for sample variability between animals, a paired t-test, Mann-Whitney rank sum test, on-way Anova and Wilcoxon test were used. Data are displayed as mean ± SEM. P values of * ≤ 0.05, ** ≤ 0.01, or *** ≤ 0.001 were considered to represent statistically significant differences.

Results

1. In vivo microinjection and electroporation of mouse testes

Two routes have been described for microinjection of DNA into the testes: direct injection through the *tunica albuginea*, or injection into the lumen of the seminiferous tubules via the *rete testis*. We chose the *rete testis* route and evaluated the efficacy of the microinjection protocol. In particular, we wished to better characterize the diffusion of the injected solution in the volume of the testis,

as we were unable to find any information on this parameter in the literature. The efficacy of microinjection *via rete testis* was assessed using fluorescent i- particles NIRFiP-180, and by measuring their diffusion in testis cross sections examined by microscopy 3 days post-injection (Fig 1). To avoid lesions due to overfilling, the injection was controlled by measuring the expansion of the staining of the peripheral seminiferous tubules during the injection. Injections were stopped when the testes were filled to 2/3 of their capacity (Fig 1A-B). In testis cross sections, the fluorescent i-particles NIRFiP-180 were heterogeneously distributed, and mainly observed in seminiferous tubules located in the peripheral region of the testes, with fewer particles present in the center of the testes (Fig 1C- D). Moreover, no fluorescent i-particles NIRFiP-180 were visible in the peritubular space. These results indicated that microinjection through the *rete testis* did not produce a homogenous distribution of the particles throughout the seminiferous tubules. Nevertheless, the seminiferous tubules remained intact, as no signal was observed in the peritubular space (Fig 1C-D).

Next, we assessed the overall safety of the *rete testis* microinjection and electroporation of mRNA and EEV into testes. The safety of the protocol was evaluated by comparing macroscopic and microscopic anatomies of control (injected with control solution, PBS, 0.05 % FG), and treated testes (injected either with EEV-*GFP* (PBS, 0.05 % FG) or *GFP*-mRNA (PBS, 0.05 % FG)). Three days post-injection, the testes were first observed under a binocular microscope to identify possible macroscopic degeneration of the seminiferous tubules (Fig 2 A1 and 2 B1). Degenerations appears as pearly white lesions at the surface of the testis as illustrated in Supp Fig 2 following over electroporation. With the protocol we have developed, no such lesions were observed. Next, the testes were measured and weighed. No statistical differences in length and weight were observed between control and treated testes (Fig 2 A2, A3, B2, B3). Then, microscopic differences were sought by histological analysis of 5 μ m sections (Fig 2C). No difference was observed between the control condition and EEV-*GFP* or *GFP*-mRNA on the full cross sections (Fig 2 C1, C2, C3). Next, we observed all the different testicular cells, including all germ cell types and Sertoli cells (Fig 2 D1, D2, D3) for each condition. The layered structure of germ cells was identical in all conditions. Analysis of the histological sections revealed no differences in the tubules area of the testes injected either with EEV-*GFP* or *GFP*-mRNA (Fig 2 E). At last, Harris-Shorr staining of the epididymal sperm cells demonstrated that there were no increases in morphological defects when mRNA and EEV were used in comparison with the control (Fig 2 F4). Taken together, these results suggest that *in vivo* microinjection and electroporation of EEV or mRNA did not perturb spermatogenesis.

2. Analysis of EEV-*GFP* and *GFP*-mRNA testicular expression by whole testis imaging

After validating the injection method, we compared the kinetics of GFP expression and the maintenance of the fluorescent signals for mRNA and EEV. To do so, we injected and electroporated one testis of adult B6D2 mice with EEV-*GFP* (n=129) or with *GFP*-mRNA (n=65). At 0-, 1-, 7-, 15-, 21-, 28-, 35-, 42- and 49-days post-injection, the whole testes were observed under an inverted microscope. The exogenous fluorescence was directly visible at the surface of the testes when illuminated with light at the appropriate wavelength (Fig 3 and Fig 4). No testicular lesions were observed on the testes at any post injection time (Fig 3 A1-H1 and Fig 4 A1-F1). In addition, both *GFP*-mRNA and EEV-*GFP* induced GFP expression in the testes (Fig 3 A2-H2 and Fig 4 A2-F2). It is worth noting that both vectors induced GFP expression at one day post-injection. However, the duration of fluorescent signals were different. For EEV, GFP fluorescence was still observable on day 42 for 100 % of samples, and 56 % of samples were positive on day 49, indicating that expression lasted around 1.5 months (Fig 4G). In contrast, for mRNA, 100 % of testes were labeled on day 21, but none showed any fluorescence on day 28 (Fig 4G). Thus, EEV transfection allowed a considerably longer duration of expression than mRNA. (Fig 3 and 4). It is important to underline that the signal measured is the fluorescence emitted by the GFP. This

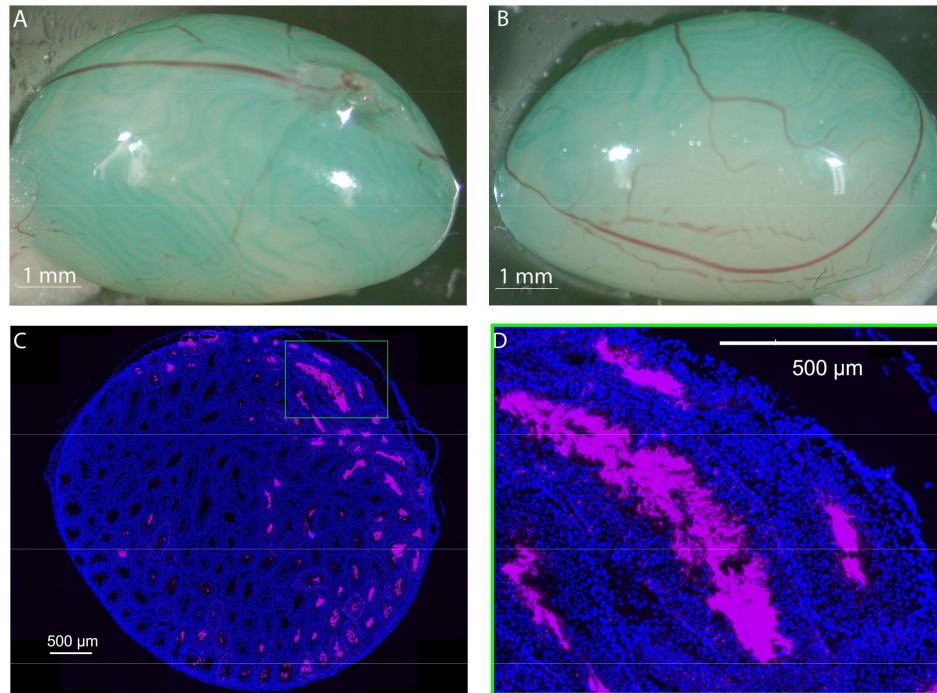


Figure 1

Distribution of I-particles NIRFIP-180 in testis injected *via* the rete testis route.

(A) A solution containing 0.05 % Fast Green and 1 % fluorescent i-particles NIRFIP-180 was prepared, 10 μ L was injected into the seminiferous tubules of adult males, through the *rete testes* and its efferent channels. Injection was performed at constant pressure under a binocular microscope. The progression of filling of the seminiferous tubules was monitored thanks to the Fast Green. (B) The testes were only filled to 2/3 capacity in order to prevent damage to the tissue. (C) Representative distribution of fluorescent i-particles NIRFIP-180 in a whole cross-section of an injected testis. Nuclei were counterstained with DAPI (blue emission) to reveal tubules. (D) Enlargement of a seminiferous tubule showing particles localized inside the lumens of the tubules. Scales bars: 1 mm and 500 μ m.

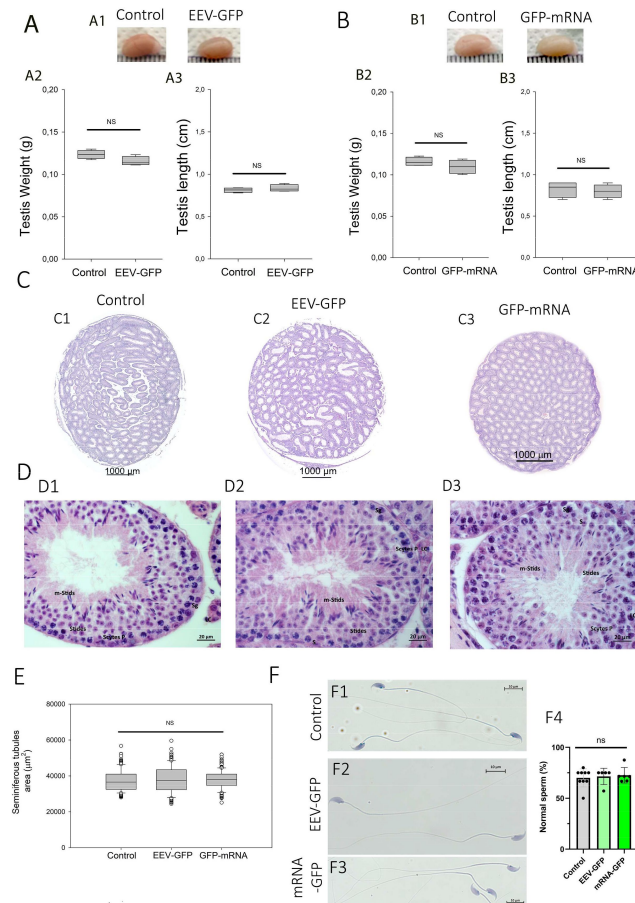


Figure 2

***In vivo* injection and electroporation do not alter the morphological structure of the testes, seminiferous tubules, or sperm cells.**

(A, B) Testicular morphology was not affected by *in vivo* injection and electroporation of EEV- GFP (A) or GFP-mRNA (B). Controls correspond to contralateral testes injected/electroporated with control solution (PBS, 0.05 % FG). (A1, B1) comparison of the testicular morphology of adult testes injected with nucleic acid vectors or control solutions. (A2, B2) Comparison of testicular weight and (A3, B3) testicular length on day 7 after injection/electroporation. Data are represented as a box plot median (n=4 for each condition). A Wilcoxon matched pairs test was used to assess the significance of any differences in testis weights and lengths, and p values of ≤ 0.05 were considered statistically significant. (C) Intact testicular structure after *in vivo* injection and electroporation with EEV-GFP and GFP-mRNA. Comparison of testicular cross section structures. Testes paraffin sections were stained with eosin/hematoxylin and observed by light microscopy (20x magnification). (C1) Control, (C2) EEV-GFP injected and (C3) GFP-mRNA injected. Scales bars: 1000 μ m. (D) Seminiferous tubule structures are not affected by *in vivo* injection and electroporation with EEV-GFP and GFP-mRNA. Enlargement of cross sections showing the fine structure of a seminiferous tubule for control (D1), EEV-GFP (D2) and GFP-mRNA (D3). In each tubule the different layers of spermatogenic cells are indicated, Sertoli cells (S), Spermatogonia (Sg), Spermatocytes (Scytes), rounds Spermatids (Stids) and sperm cells (Spz), Leydig cells (L). Scales bars: 20 μ m. (E) The area of seminiferous tubules is not affected by *in vivo* injection and electroporation with EEV-GFP and GFP-mRNA. Comparison of the seminiferous tubule diameter after injection of nucleic acid vectors or control solutions. Data are represented as a box plot median. The areas of seminiferous tubules (μ m²) were measured for round cross sections of $n > 35$ tubules per testis section (n= 5 testis sections per condition). Statistical significance was verified using a Student's *t*-test. (F) Injection/electroporation do not impact epididymal sperm cells. Representative sperm observed by light microscopy on day 7 after injection/electroporation with Control solution (F1), EEV-GFP (F2), or GFP-mRNA (F3). Scale bars: 10 μ m. (F4) Percentage of normal epididymal sperm cells in each condition. The number of males were n=5 for EEV-GFP; n=6 for GFP-mRNA and n=9 for WT. More than 150 sperm by males were analyzed. Statistical significance was verified using a one-way ANOVA test.

signal is dependent on both the half-lives of the plasmid/mRNA and the GFP. Therefore, the kinetic of the signal persistence (which is called here expression) is a combination of the persistence of the vector and the synthesized protein. In addition to differences in duration of expression, the GFP expression patterns were clearly different: mRNA produced a large, diffuse pattern, highlighting the shape of the seminiferous tubules; EEV-GFP produced a punctiform pattern (**Fig 3B** and **4B**.) These results suggest that GFP-mRNA and EEV-GFP targeted different seminiferous cell types, such as Sertoli cells and all germline cells, or that there were differences in terms of transfection efficiency.

3. Kinetics of EEV and mRNA expression assessed by *in vivo* imaging

To further assess and compare the kinetics of the expression of the two vectors, we expressed exogenous luciferase in the testis using EEV or mRNA and observed the level of luciferase by *in vivo* bioluminescence imaging. For EEV, we took advantage of the fact that the EEV plasmid contains the DNA sequence of the luciferase protein (*CAGs-GFP-T2A-Luciferase*) in addition to the DNA sequence of the GFP fluorescent protein (Supp **Fig 1**). For mRNA, we injected commercial naked *luciferase*-mRNA into the *rete testis* according to the same protocol as used for GFP-mRNA. For this set of experiments, we injected the EEV-GFP-Luc and *luciferase*-mRNA into the testes of 6 adult mice on day 0. We injected a similar number of copies of mRNA and EEV. The testes were imaged *in vivo* to detect bioluminescence expression, on day 1 and weekly until disappearance of the signal, no more than 120 days post-injection (**Fig 5**). For EEV-GFP-Luciferase, the bioluminescence induced by transfection was detected from day 1. After a rapid decrease in signal intensity over the first 3 weeks, a weak but constant signal remained detectable for 3 months, then faded away (**Fig 5 A1-A2**). For *Luciferase*-mRNA, expression was also detected from day 1. The bioluminescence decreased gradually over 3 weeks, becoming undetectable thereafter (**Fig 5B1-B2**). These results are consistent with our previous results (**Fig 3**) and confirm that EEV allows a longer expression of exogenous protein within the testis. **Fig 5C** compare the kinetics of expression observed with EEV and mRNA. Overall, our results indicated a stronger expression for mRNA than for EEV, but with expression decreasing rapidly over-time, and almost no remaining signal after 3 weeks. In contrast, residual expression was detectable for several months with EEV.

4. Assessing testicular and cellular GFP-mRNA expression using whole testicle optical clearing, lightsheet microscopy, and 3D reconstructions

To better characterize the spatial distribution of GFP-mRNA expression in the testis, we performed whole testicular optical clearing. On day 0, we injected and electroporated testes with GFP-mRNA (n=6). On day 1, we harvested the testes and cleared them with a modified optical clearing protocol, as described in the MM section. After complete clearing (**Fig 6 A**), we imaged the whole testes using a lightsheet microscope and performed 3D reconstruction from the images stack obtained (videos 1, 2 and **Fig 6 B**). From this 3D reconstruction, we determined the volume of the testis stained with GFP by measuring the GFP-positive area in each image and multiplying it by the thickness of the z-step (10 μm). Due to optical issues, only half part of the whole testis was acquired, then the sample was turned by 180° to acquire the other half part. A total GFP-stained volume of 0.51 mm^3 and 0.23 mm^3 was determined from the face A and the face B, respectively. The corresponding total volume of half part of the testis was measured as 60 mm^3 , therefore 0.81 % and 0.24 % of the testis was transfected for the face A and the face B, respectively (**Fig 6 B**).

EEV-GFP

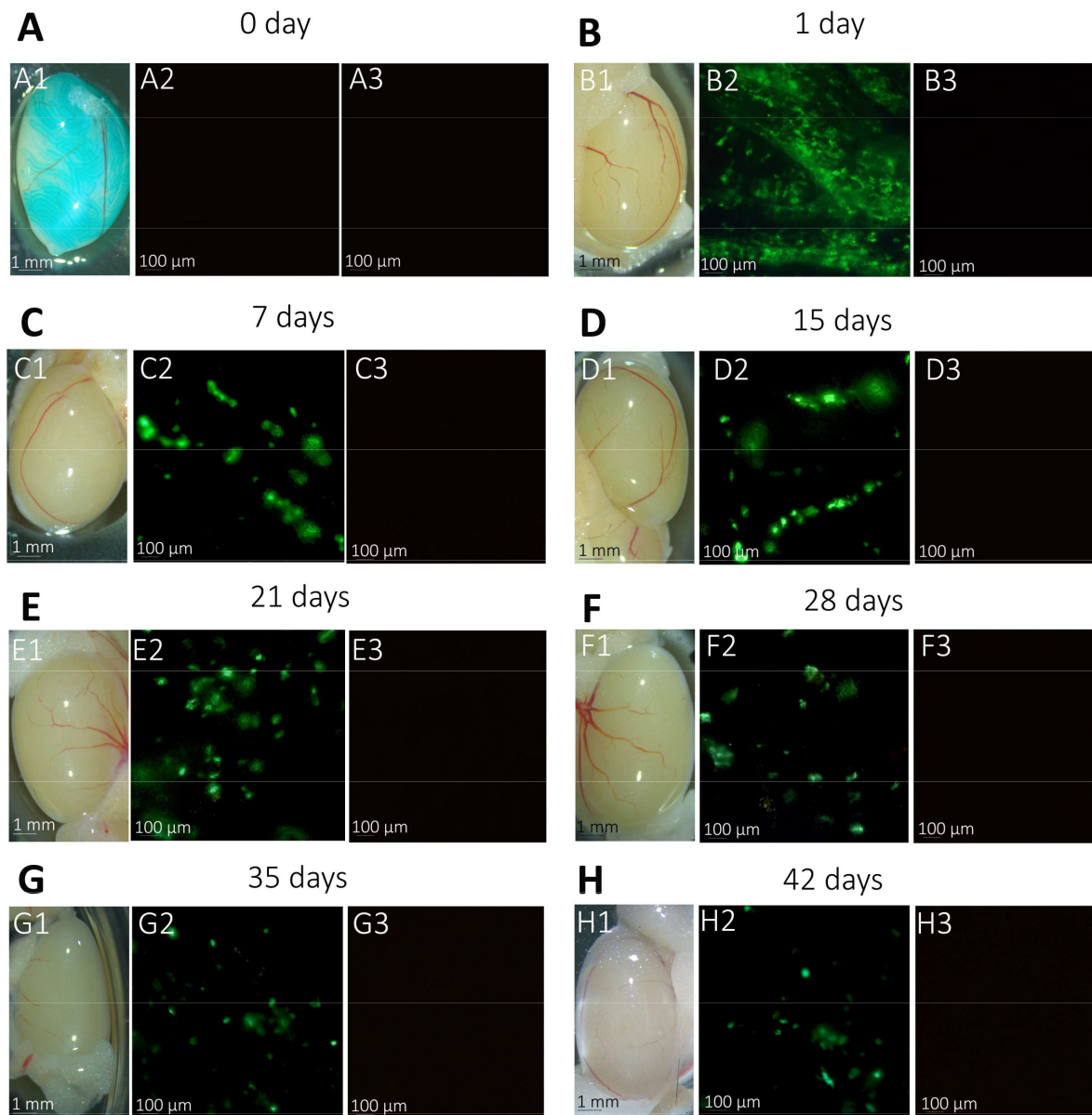


Figure 3

Kinetics of EEV-GFP expression following *in vivo* injection/electroporation: whole testicular expression

(A1-H1) Whole-mount testes on days 0, 1, 7, 14, 21, 28, 35 and 42 after *in vivo* injection/electroporation with EEV-GFP. (A2-H2) Under fluorescence observation, GFP expression was detectable in transfected testes from 12-week-old B6D2 mice. (C3-H3) Insets show the absence of autofluorescence in non-transfected control testes, observed under 4X magnification. The GFP expression presented a punctiform pattern in seminiferous tubules and was detected from 1 day to 42 days. Scales bars: 1 mm and 100 μm.

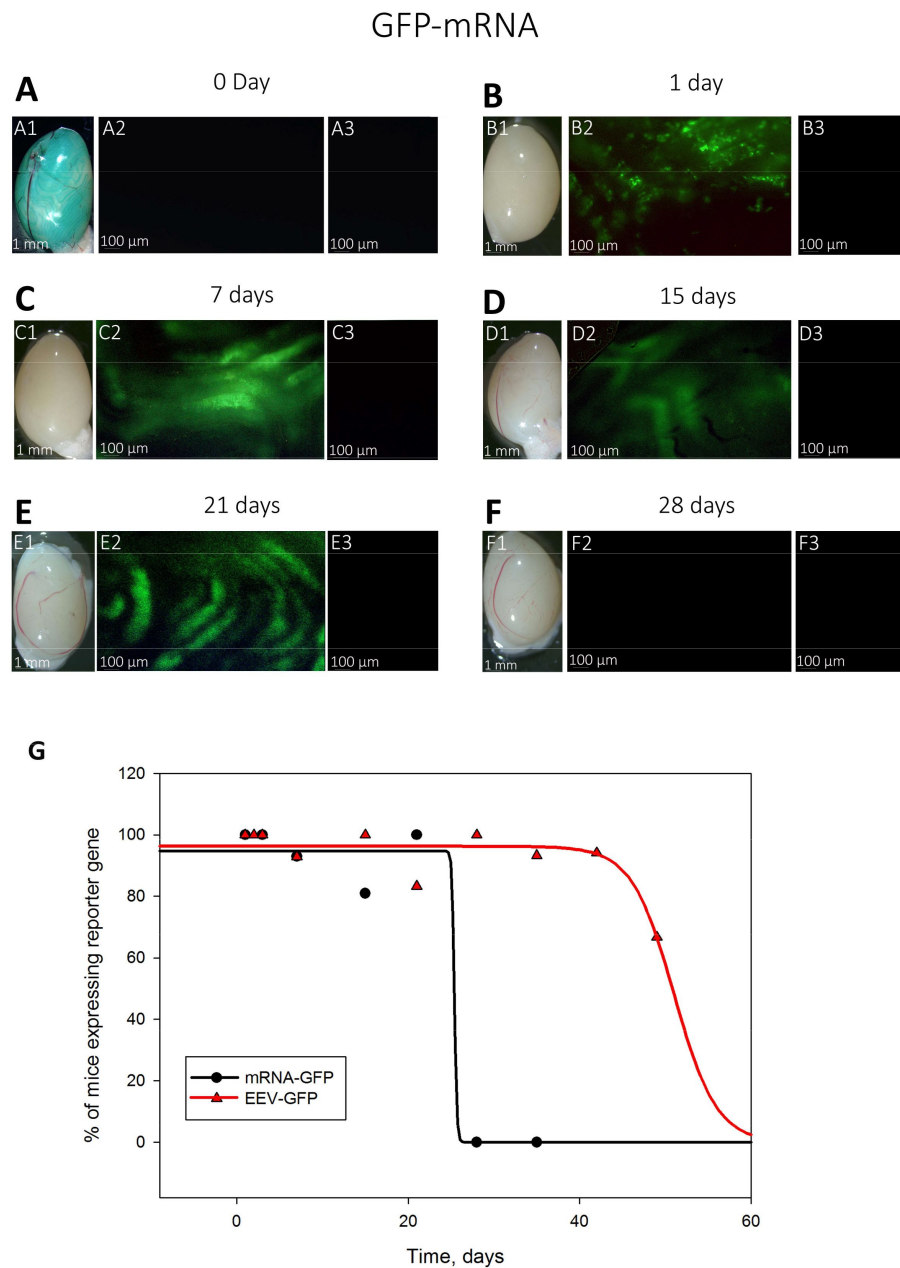


Figure 4

Kinetics of *GFP*-mRNA expression following *in vivo* injection/electroporation: whole testicular expression

(A1-F1) Whole-mount testes on days 0, 1, 7, 15, 21 and 28 after *in vivo* injection/ electroporation with *GFP*-mRNA. (A2-F2) Under fluorescence observation, GFP expression was detectable in transfected testes from 12-week-old B6D2 mice. (A3-F3) Insets show the absence of autofluorescence in non-transfected control testes, observed under 4X magnification. The GFP expression presented a continuous pattern in seminiferous tubules and was detected from day 1 to day 15. Scale bars: 1 mm and 100 μ m.

(G) Comparison of the percentage of injected mice exhibiting reporter gene expression. Mice injected with *GFP*-mRNA exhibited GFP expression from day 1 to day 21. Mice injected with EEV-GFP exhibited GFP expression from day 1 to day 49. (for EEV-GFP n=11 on day 1; n=13 on day 2; n=10 on day 3; n=14 on day 7; n= 5 on day 10; n= 12 on day 15; n=11 on day 21; n= 12 on day 28; n=15 on day 35; n=17 on day 42 and n=9 on day 49) ; (for *GFP*-mRNA n=3 on day 1; n=4 on day 3; n=15 on day 7; n= 21 on day 15; n=15 on day 21 and n= 5 on day 28).

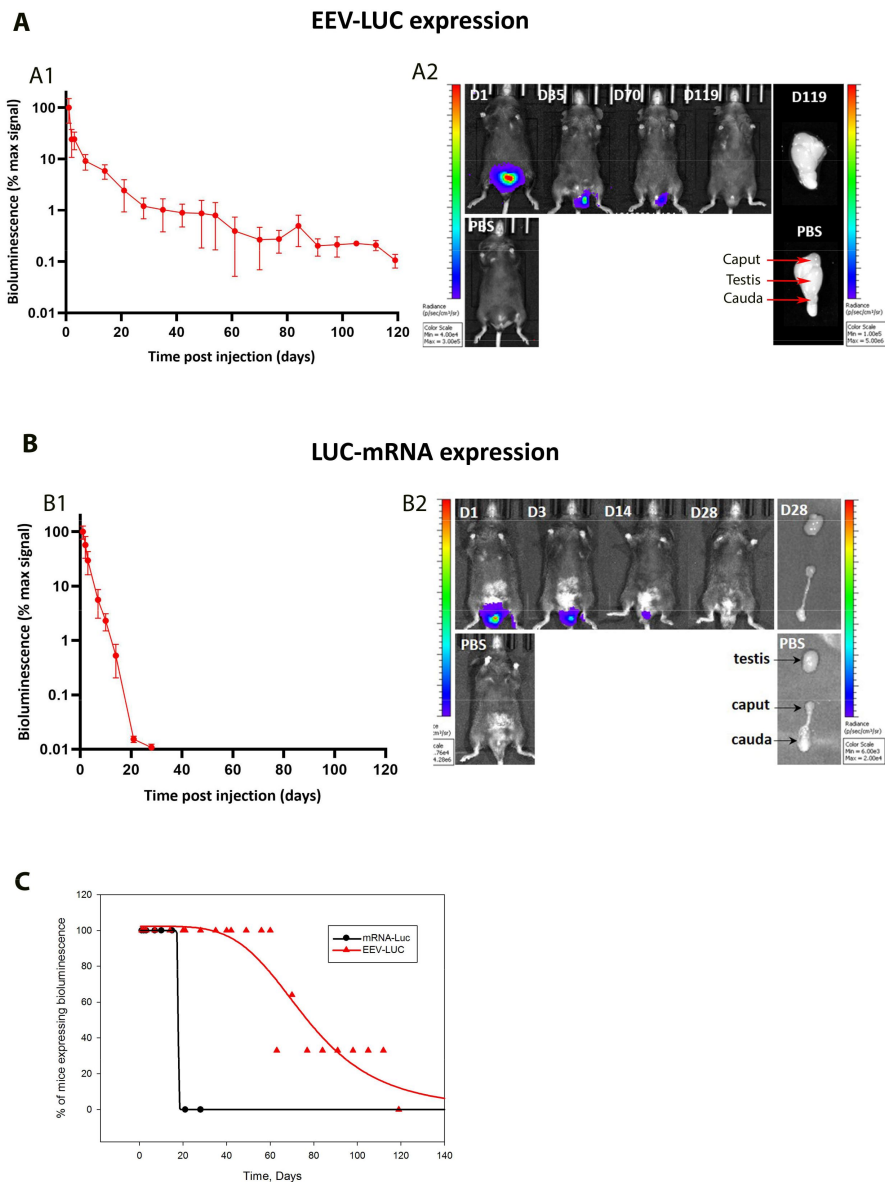


Figure 5

Kinetics of EEV and mRNA expression by *in vivo* bioluminescence imaging.

(A) *In vivo* bioluminescence imaging quantification of luciferase expression over time following injection/electroporation of EEV-GFP-luc. (A1) EEV-GFP-Luc was injected into the testes and electroporated on day 0. Bioluminescence signal was quantified at several time points. Results are expressed as a percentage of the maximal signal (mean $\pm$ SEM; n=5 mice up to D2; n=4 from D3 to D28; n=3 from D35 to D98 and n=3 from D105 to D119). (A2) *In vivo* bioluminescence images of a representative mouse at several time points after administering EEV-GFP-LUC or PBS, and ex vivo bioluminescence images of testes after 119 days.

(B) *In vivo* bioluminescence imaging quantification of luciferase expression over time induced by injection/electroporation of LUC-mRNA. (B1) LUC-mRNA was injected into the testes and electroporated on day 0. Bioluminescence signal was quantified in the whole testis at several time points. Results are expressed as a percentage of the maximal signal (mean $\pm$ SEM; n = 5 mice). (B2) *In vivo* bioluminescence images of a representative mouse at several time points after administering LUC-mRNA or PBS, and ex vivo bioluminescence images of caput, testes, and cauda after 28 days.

(C) Decay over time of the number of mice expressing reporter genes. Mice were injected on day 0 with LUC-mRNA or EEV-GFP-LUC and the number of mice showing bioluminescence in the testis was counted at different time points, from day 1 to day 119. For EEV-GFP: n=13 at D1; n=13 at D2; n=4 from D3 to D28; n=3 from D35 to D98 and n=3 from D105 to D119. For LUC-mRNA: n = 5 mice for all-time points.

GFP-mRNA 1 day post injection

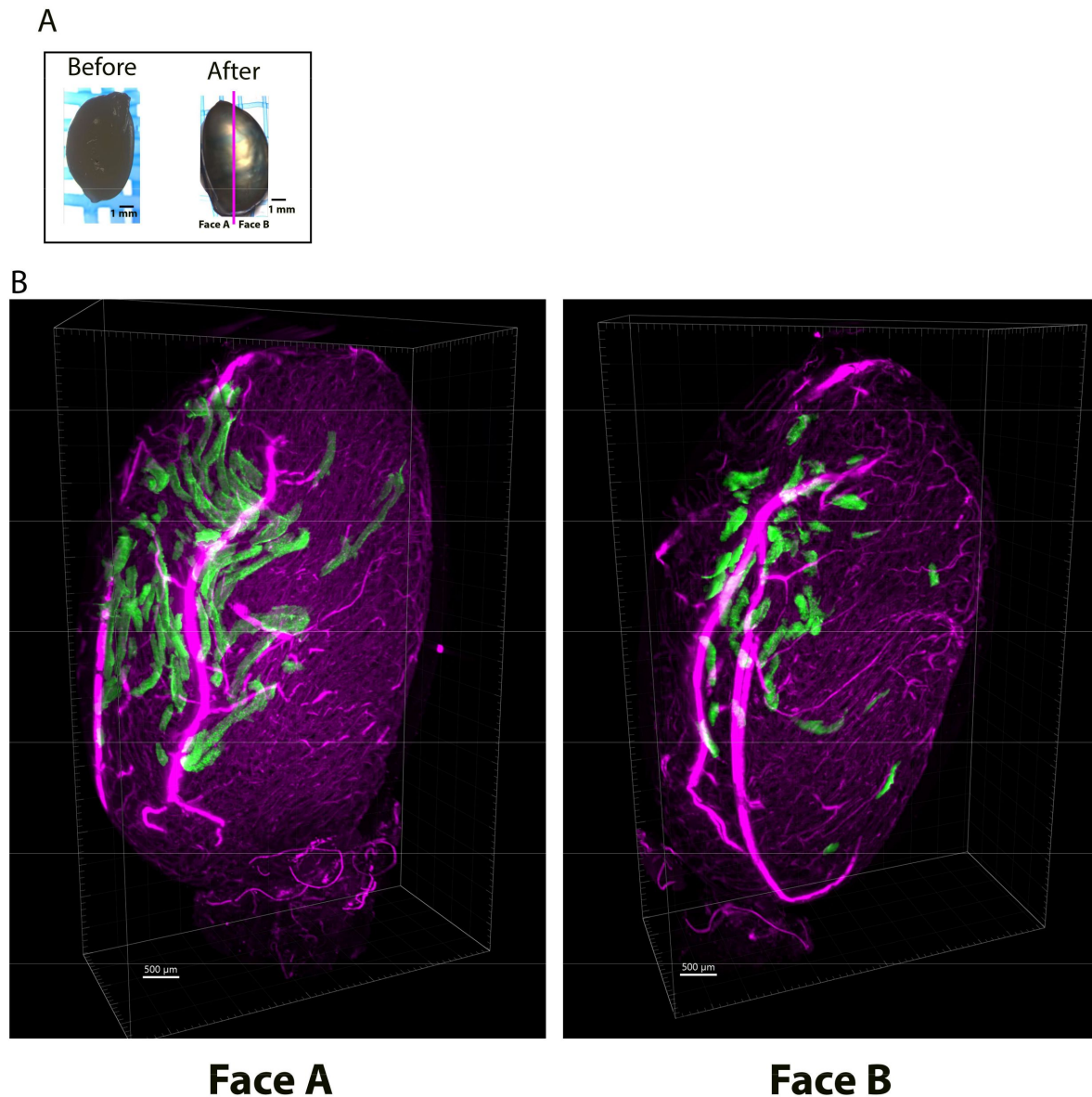


Figure 6

Testicular and cellular *GFP*-mRNA expression measured on optically cleared testis after 3D image reconstructions from lightsheet microscopy imaging.

Testes were injected/electroporated with *GFP*-mRNA on day 0. On day 1, whole testes were fixed and subjected to optical clearing. (A) Testes were observed before and after optical clearing on a binocular microscope. The right image shows the transparency of the testis after complete clearing, revealing the blue mesh throughout the organ. (B) The 3D internal structure of a cleared testis was reconstructed from the lightsheet microscopy images. The reconstruction was possible only for a half testis due to optical issues. Two opposing faces of the same testis are presented, allowing the distribution of *GFP* fluorescence throughout the seminiferous tubules to be measured. Pink fluorescence corresponds to the autofluorescence of interstitial cells located around the seminiferous tubules. Scale bars A: 1 mm and B: 500 μ m.

5. Assessing GFP cellular expression using whole testicle optical clearing and adaptive optics confocal microscopy

Because the GFP fluorescence patterns were different for the two nucleic vectors when observed under the inverted microscope (**Fig 3**), we wondered whether the same cell types were targeted in both cases. To address this question, the whole optical cleared testes from EEV-GFP and GFP-mRNA-transfected mice were imaged using a confocal microscope. The different cell types were identified based on their positions within the seminiferous tubule, their cellular shape and their nuclear morphology - revealed by nuclear staining. For instance, Sertoli cells have an oval to elongated nucleus and the cytoplasm presents a complex shape (“tombstone” pattern) along the basement membrane [27]. Round spermatids have small, round and compact nuclei with a nucleolus and are localized between the spermatocytes and elongated spermatids [28]. For EEV-GFP, on day 1 post injection and electroporation, a strong punctiform green fluorescence was visible inside the seminiferous tubules (**Fig 7 A**). Based on the different morphological criteria, this fluorescent signal was detected in spermatocytes, round spermatids and Sertoli cells (**Fig 7 BC**). On day 7, the GFP signal induced by EEV-GFP was reduced and only isolated signals in a few seminiferous tubules (1 per 11 tubules) were observed (**Fig 7 D**). These signals were associated only with Sertoli cells only (**Fig 7 E**).

For the mRNA vector, on day 1 and day 7 post-injection and electroporation, an intense fluorescence was observed in all the germ cells and in Sertoli cells in the seminiferous tubules (**Fig 8A**). At the cellular level, this fluorescent signal was associated with spermatogonia, spermatocytes, round spermatids, elongated spermatids, and mature spermatids cells to similar extents for both post-injection times (**Fig 8 B, D**). In contrast to what was observed with EEV on day 7, no reduction in the number of fluorescent seminiferous tubules was noted when using GFP-mRNA, with 8 out of 10 tubules stained (**Fig 8 CD**).

6. Expression of naked *mCherry*-mRNA in testis following electroporation

Heterologous expression is well known to depend on the protein studied, we therefore tested the same reporter proteins to compare the mRNA and EEV vectors in the experiments presented above. Apart from the bioluminescence experiments with luciferase, we compared only GFP protein expression. To validate and confirm the capacity of naked mRNA to express proteins in the testes after injection and electroporation, we further challenged the method with mCherry, another reporter protein (**Supp Fig 1BD**). We injected homemade naked mRNA coding for mCherry into the testes. As previously with GFP-mRNA, no testicular lesions were observed (**Supp Fig 3** A1, B1, C1, D1, E1, F1).

The assessment of the temporal persistence of testicular mCherry fluorescent protein expression revealed a robust red fluorescence from day 1 post-injection, which remained detectable for at least 15 days (**Supp Fig 3** B2, C2, and D2). At the cellular level, the fluorescent signal was detected in germ cells, including spermatogonia, spermatocytes, round spermatids, mature spermatids, and Sertoli cells on days 1 and 7 post-injection (**Supp Fig 4**).

Finally, we compared the kinetics and levels of expression of the three different mRNA molecules, coding for mCherry, GFP and luciferase. By comparing the number of mice expressing *mCherry*-mRNA, *GFP*-mRNA and *luciferase*-mRNA fluorescence/luminescence over 21 days, we observed first that expression was detectable for all mRNAs on day 1, and second that the duration of expression was slightly different for the individual mRNAs. For instance, on day 15, 100 %, 80 %

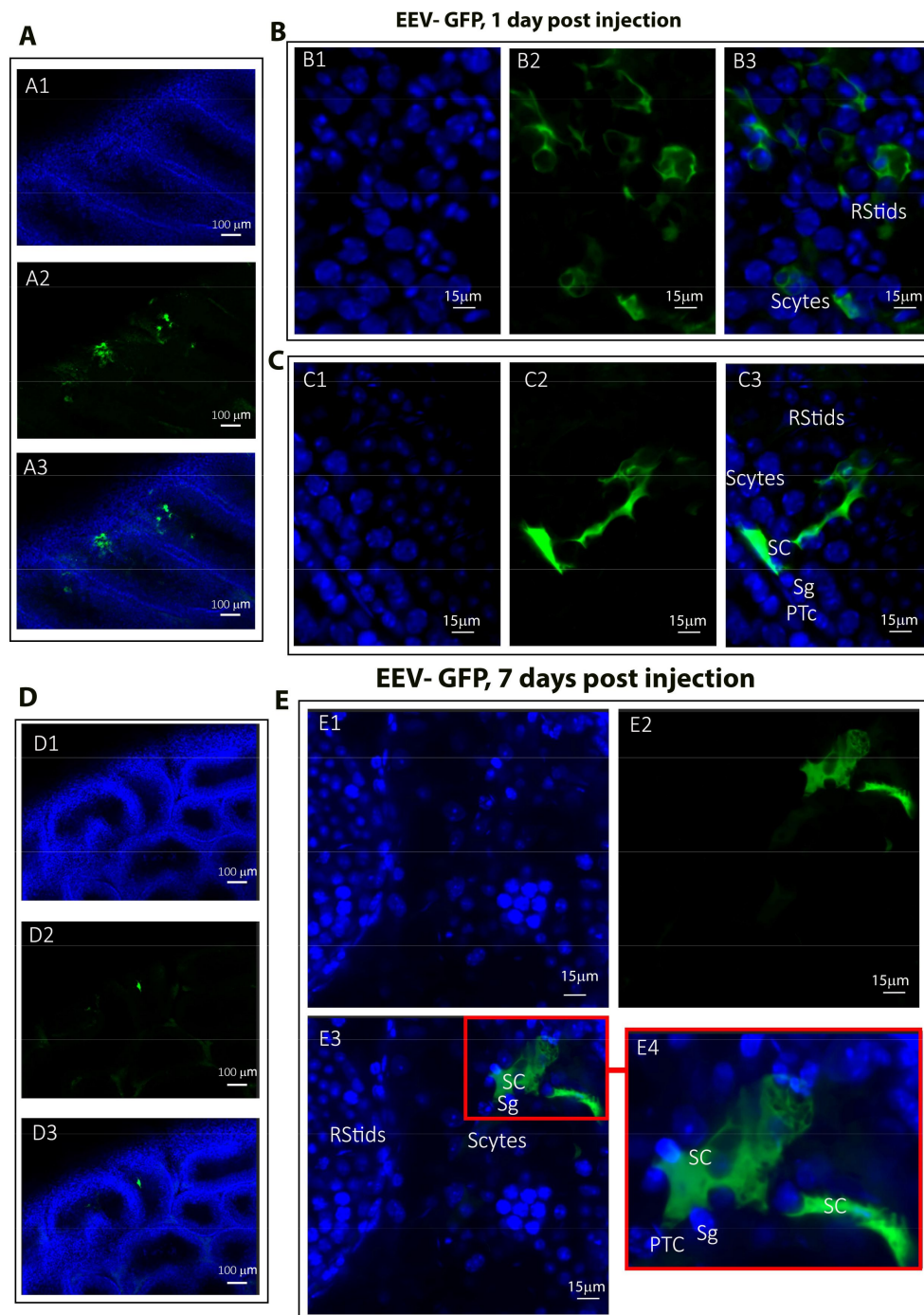


Figure 7

Cellular expression of EEV-GFP following *in vivo* injection/ electroporation.

Testes were injected/electroporated with EEV-GFP on day 0. On day 1 and on day 7, whole testes were fixed and subjected to optical clearing. Cleared testes were observed by fluorescence microscopy. (A1-A3) On day 1, transfected seminiferous tubules showed dotted green fluorescence at low magnification (10x/0.45). Nuclei were counterstained with DAPI (blue staining) to reveal the structure of the seminiferous tubules. At the cellular level, fluorescence was detectable (B1-B3) in germ cells including Spermatogonia (Sg), Spermatocytes (Scytes) and round Spermatids (RStids), as well as (C1-C3) in Sertoli cells (SC). (D1-D3) On day 7, the GFP signal was lower at low magnification (10x/0.45) and detectable (E1-E3) only in Sertoli cells (40x/1.15 WI) (n=3) (PTC = Peri-tubular myoid cell). E4 is an enlargement of the red square in E3, allowing the cell type to be identified. Scale bars: 100 μm, 15 μm and 3 μm.

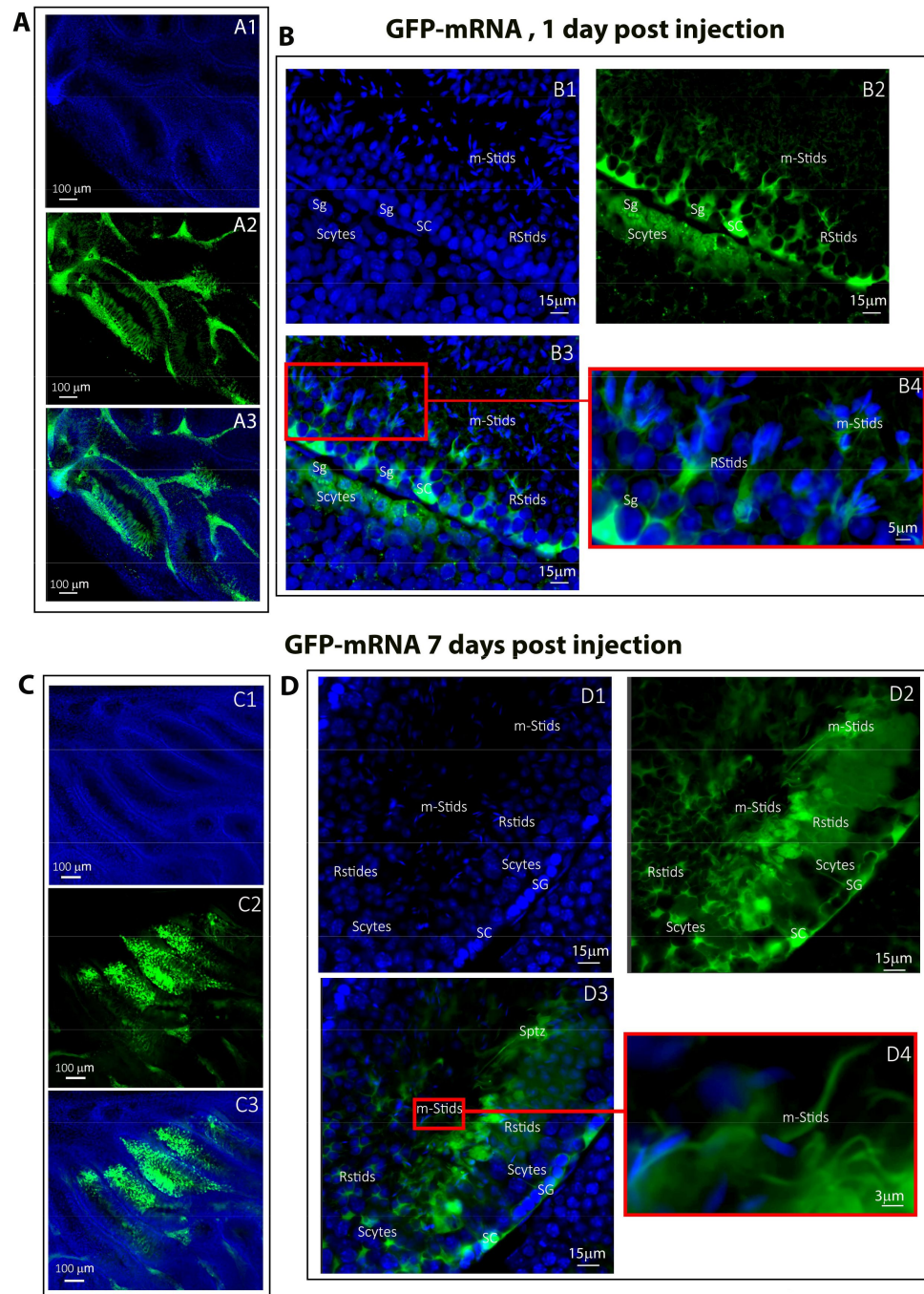


Figure 8

Cellular expression of *GFP*-mRNA following *in vivo* injection/electroporation.

Testes were injected/electroporated with *GFP*-mRNA on day 0. On day 1 and day 7, whole testes were fixed and subjected to optical clearing. Cleared testes were observed by fluorescence microscopy. (A1-A3) On day 1, transfected seminiferous tubules showed strong broad-ranging green fluorescence at low magnification (10x/0.45). Nuclei were counterstained with DAPI (blue staining) to reveal the structure of the seminiferous tubule. At the cellular level, fluorescence was detectable in germ cells (B1-B3) including Spermatogonia (Sg), Spermatocytes (Scytes) and round Spermatids (RStids), mature spermatids cells (m-Sptids) and Sertoli cells (SC). B4 is an enlargement of the red square in B3, allowing the cell types to be identified. (D1-D3) On day 7, the GFP signal remained strong at low magnification (10x/0.45) and was still detectable in (E1-E3) all germ cell types and Sertoli cells (40x/1.15 WI) (n=3). E4 is an enlargement of the red square in E3, showing that testicular sperm were also stained. Scale bars: 100 μ m, 15 μ m and 3 μ m.

and 60 % of mice injected with *GFP*-mRNA, *mCherry*-mRNA, and mRNA- *luciferase*, respectively presented fluorescence/bioluminescence and on day 21, 100 % of mice expressed GFP, whereas no signal was observed for mCherry or Luciferase (**Supp Fig 5** [↗](#)).

7. Endogenous expression of ARMC2 in germ cells

Before attempting to rescue expression, we felt it was important to better characterize *Armc2* expression in healthy germ cells, and in particular to study the timing of expression.

IF was used to determine when ARMC2 protein was detectable during spermatogenesis. For these experiments, dissociated cells from testes were observed to detect the presence of ARMC2 on different spermatogenic cells. ARMC2 was present only in the flagella of the elongated spermatids (**Fig 9A** [↗](#) and **Supp Fig 6A** [↗](#)). The specificity of the signal was validated using testicular cells from *Armc2* KO mice, where no signal was observed on all spermatogenic cells (**Fig. 9B** [↗](#)). In transversal sections of WT seminiferous tubules, ARMC2 signal was not present in spermatogonia and spermatocytes (**Supp Fig 6B** [↗](#)), but detected in spermatid layers.

By analyzing the RNA-seq database produced by Gan's team [29 [↗](#)], we show that the mRNA encoding ARMC2 starts to be expressed at the pachytene spermatocyte stage, then shows a gradual increase at the round spermatid stage, and finally becomes predominantly expressed at the elongated spermatid stage (**Supp Fig 6C** [↗](#)), a result in agreement with a post-meiotic function of the protein. Finally, we also consistently observed a staining at the base of the manchette of elongating spermatids but we have no explanation for that (**Fig 9A3** [↗](#)).

In conclusion, the results presented here demonstrate that the ARMC2 protein is expressed and translated at the late stages of spermatogenesis.

8. Co-injection of *Armc2*-mRNA and *eGFP*-mRNA followed by electroporation is safe and induces green fluorescence in the seminiferous tubules

We next tested whether the injection and electroporation of *Armc2*-mRNA molecules had any deleterious effects on testis anatomy and seminiferous tubule structure. We first verified the quality of *Armc2*-mRNA synthesis by transfecting HEK cells and performing Western blot (**Supp Fig 7** [↗](#)). After this validation, we co-injected *Armc2*-mRNA and *eGFP*-mRNA into the left testes of mice, using the right testes as untreated controls. *eGFP*-mRNA was co-injected to verify and monitor transfection efficiency. The testes were observed under a binocular microscope at different times (3-, 6-, 10-, 15-, 21-, 28- and 35-days) after electroporation to identify possible macroscopic degeneration of the seminiferous tubules. No morphological defects were observed in the testes co-injected with *Armc2*-mRNA and *eGFP*-mRNA. An example of control and injected testes from day 15 is presented in **Fig 10** [↗](#) A1, B1. The testes were also weighed at different times post-injection, and the weight ratios of injected testes to non-injected control testes were determined. For all-time points, this ratio was close to 1 (**Fig 10 C** [↗](#)), confirming that the method and the mRNAs did not cause any injury at the organ level. Next, under blue light, the efficiency of the transfection was assessed by observing the GFP fluorescence at the surface of the testes. GFP fluorescence was observed on testes injected with *Armc2*-mRNA and *eGFP*-mRNA 2 weeks after injection (**Fig 10** [↗](#) B2), indicating that the naked mRNA was successfully transfected into testicular cells.

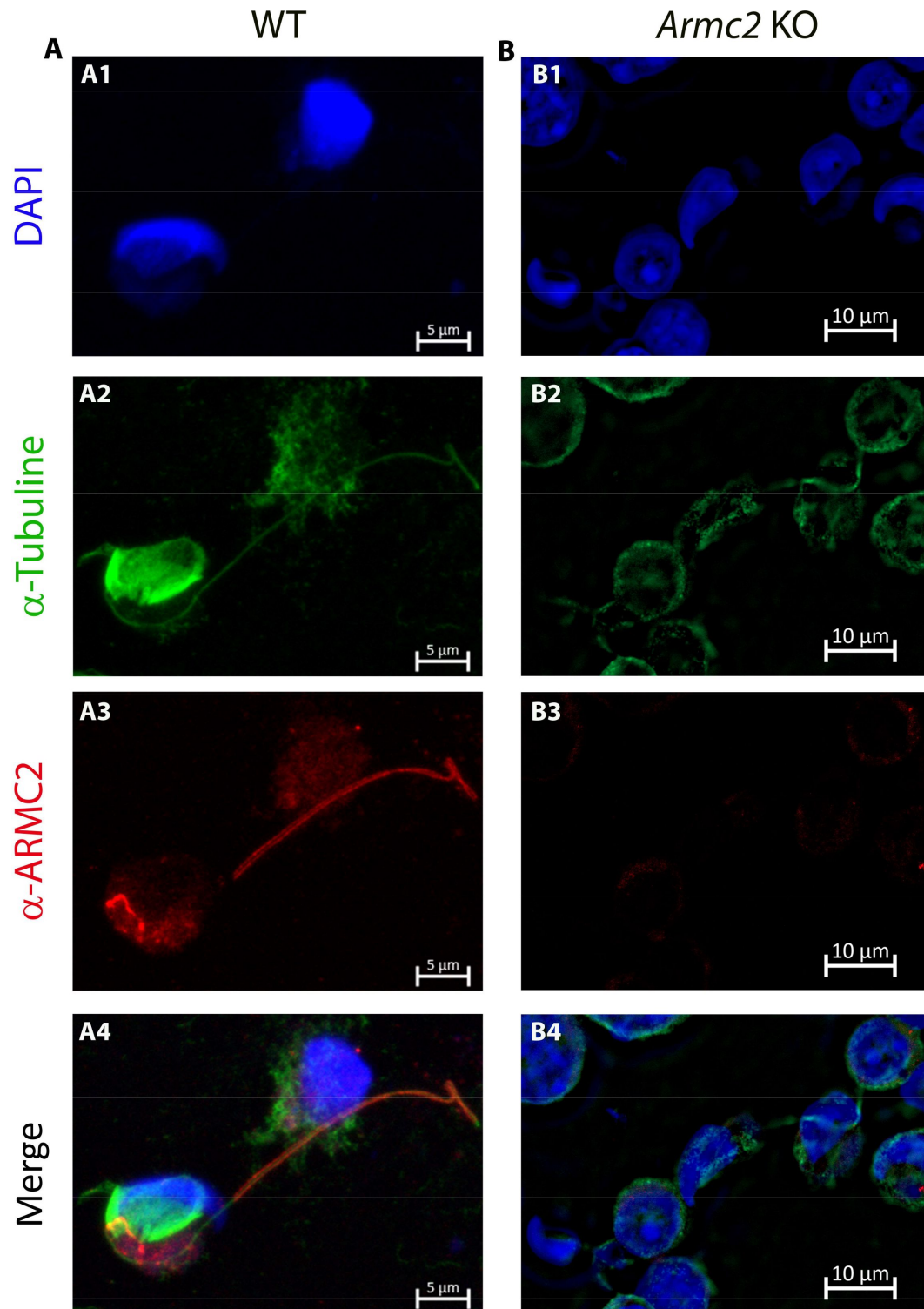


Figure 9

ARMC2 localization in dissociated testicular cells observed by immunofluorescence.

Cells from WT and *Armc2* KO mice were counterstained with Hoechst (A1-B1) and stained with antibodies against tubulin (A2-B2, green signal) and ARMC2 (A3-B3, red signal). (A4-B4) overlay of the different staining. In WT mice, ARMC2 is located in the flagellum of spermatids. In KO mice, no ARMC2 signal (red fluorescence) was observed in any cells.

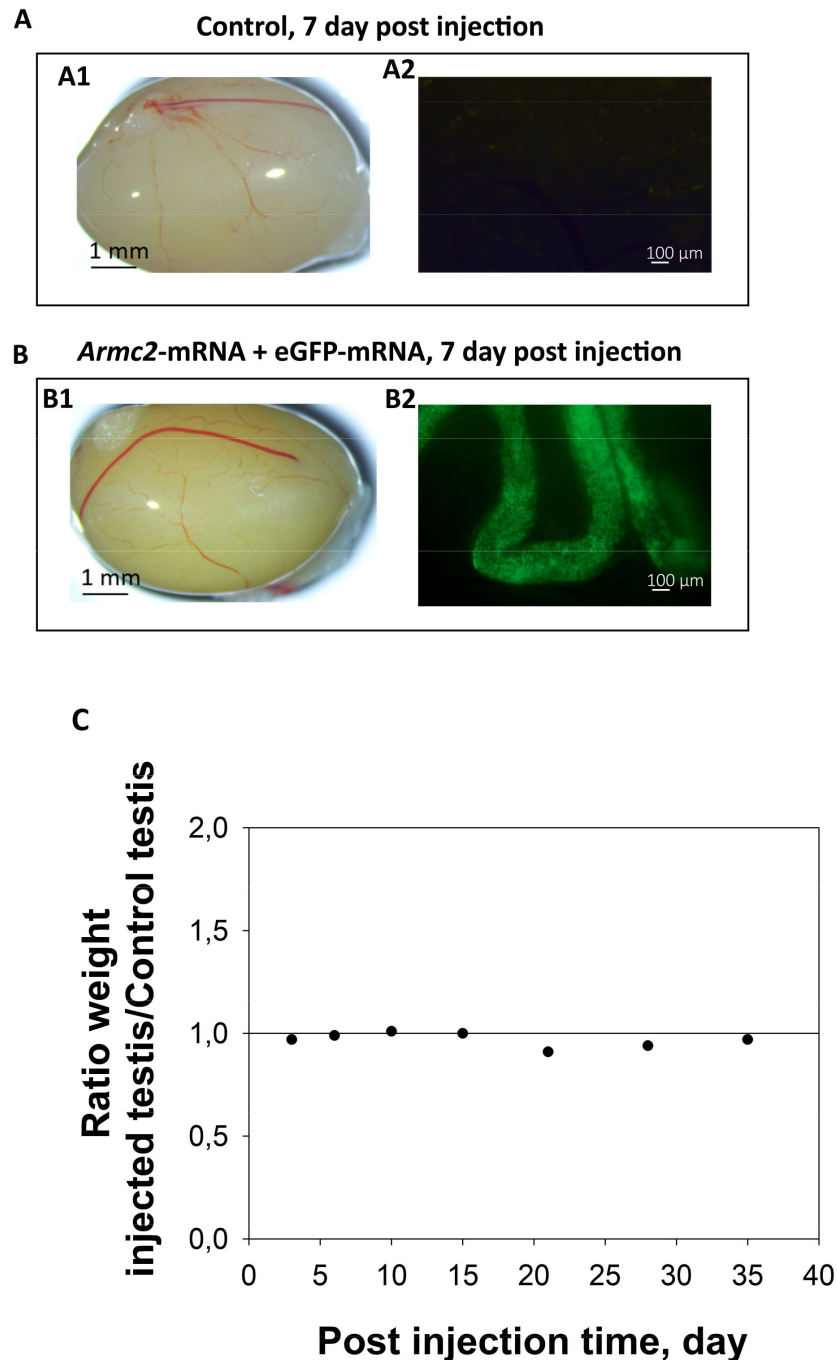


Figure 10

***In vivo* co-injection of *Armc2*-mRNA and *eGFP*-mRNA followed by electroporation do not affect testes morphology and weight.**

Adult WT mouse testes were injected with a solution containing *Armc2*-mRNA and *eGFP*-mRNA. After injection, the testes were electroporated and mice were euthanized two weeks later. (A): Whole testis under white and blue lights on a fluorescence microscope (A1): control testes not injected/electroporated (A2): testes injected with *Armc2*-mRNA and *eGFP*-mRNA. *eGFP*-mRNA was co-injected to follow the transfection efficiency. (B): Ratio of injected/electroporated testis weights to control testis weights at several time points post-injection (3-, 6-, 10-, 15-, 21-, 28- and 35-days post- surgery). n= 1 mouse per time.

9. Motile sperm cells detected in *Armc2* KO mice following *Armc2*-mRNA injection and electroporation into testes

We then assessed whether the injection of *Armc2*-mRNA into the testes in *Armc2* KO mice restored sperm motility. We examined motility of sperm cells present in the caudal part of the epididymis at different times post-injection (3- to 35-days post-injection). For each condition, between 3 and 16 KO mice were used.

The *Armc2* KO model used is known to produce sperm cells with short and irregular flagella that are therefore immotile on day 0. No motile sperm were observed on days 3, 6, 10, 15 or 28 after surgery (**Fig 11 A**). However, motile sperm cells were found in the epididymis of some *Armc2* KO mice at 21- and 35-days post-treatment (**Fig 11 A**). Indeed, 1 in 3 mice had motile sperm at 21 days post-surgery, rising to 3 in 4 mice at 35 days post-injection. Nevertheless, the number of motile sperm observed remained low: 5.5 % after 21-days and 7.15 % after 35- days post-injection (**Fig 11 A1**). The sperm motility parameters of *Armc2*^{-/-}-rescued motile sperm were characterized in comparison to those of *Armc2*^{+/+} sperm using the computer- assisted semen analysis (CASA) system (**Fig 11 A2**). These parameters included VAP, VSL, VCL, ALH, BCF, and STR. We have observed significant differences between WT and rescued sperm. In particular, the VSL and LIN parameters are lower for rescued sperm. Next sperm were sorted as progressive, intermediate, hyperactivated, or slow according to motility parameters of motile sperm and recorded from their track (**Figure 11A3**). The percentage of hyperactivated sperm and the proportion of intermediates in the *Armc2*^{-/-}-rescued motile sperm population were found to be increased in comparison to the control. Videos showing sperm motility in different conditions are available in the online material associated with this article (Videos 3 to 6).

After verifying motility, we looked at the morphology of the spermatozoa present in the cauda epididymis. Six days after injection of *Armc2*-mRNA, the cells detected were mostly round cells and abnormal spermatozoa with a short or coiled flagellum measuring between 7 and 20 μ m.

The same cell types were observed at 3-, 10-, 15- and 28-days post-surgery. In contrast, the motile sperm detected on days 21 and 35, had a normal morphology with a long flagellum (greater than 100 μ m) and a hook-shaped head (**Fig 11 B** and **Supp Fig 8**).

10. Motile sperm cells detected in *Armc2* KO mice following *Armc2*-mRNA injection and electroporation into testes are fertile

We subsequently evaluated the efficacy of *Armc2*-mRNA injection into the testes of *Armc2* KO mice in restoring sperm fertility. The fertility outcome was assessed through *in vitro* fertilization (IVF) and intracytoplasmic sperm injection (ICSI) experiments. The sperm rescued from *Armc2*^{-/-} mice were capable of successfully fertilizing eggs and producing embryos at the two-cell stage by IVF (**Fig 12 A-B-C**). Notably, 62.7% of two-cell embryos were obtained with the *Armc2*^{-/-}-rescued sperm by IVF, compared to only 2.67% with the *Armc2*^{-/-} sperm. Three % of eggs became 2-cell embryos when fertilized with sperm from *Armc2*^{-/-}, a rate not significantly different to that observed for eggs incubated 24 h without sperm, and likely corresponding to parthenogenesis activation (**Fig 12 A1-A2-B**).

To gain further insight, a comparative analysis of the developmental outcomes of mouse embryos generated by ICSI with spermatozoa from wild-type (WT), *Armc2*^{-/-} and *Armc2*^{-/-} treated mice was performed. It should be noted that in case of *Armc2*^{-/-} treated mice, the ICSI procedure was performed only with the motile *Armc2*^{-/-}-rescue spermatozoa. The percentage of live injected oocytes that reached the blastocyst embryos was 46% for WT spermatozoa, 25% for *Armc2*^{-/-}-

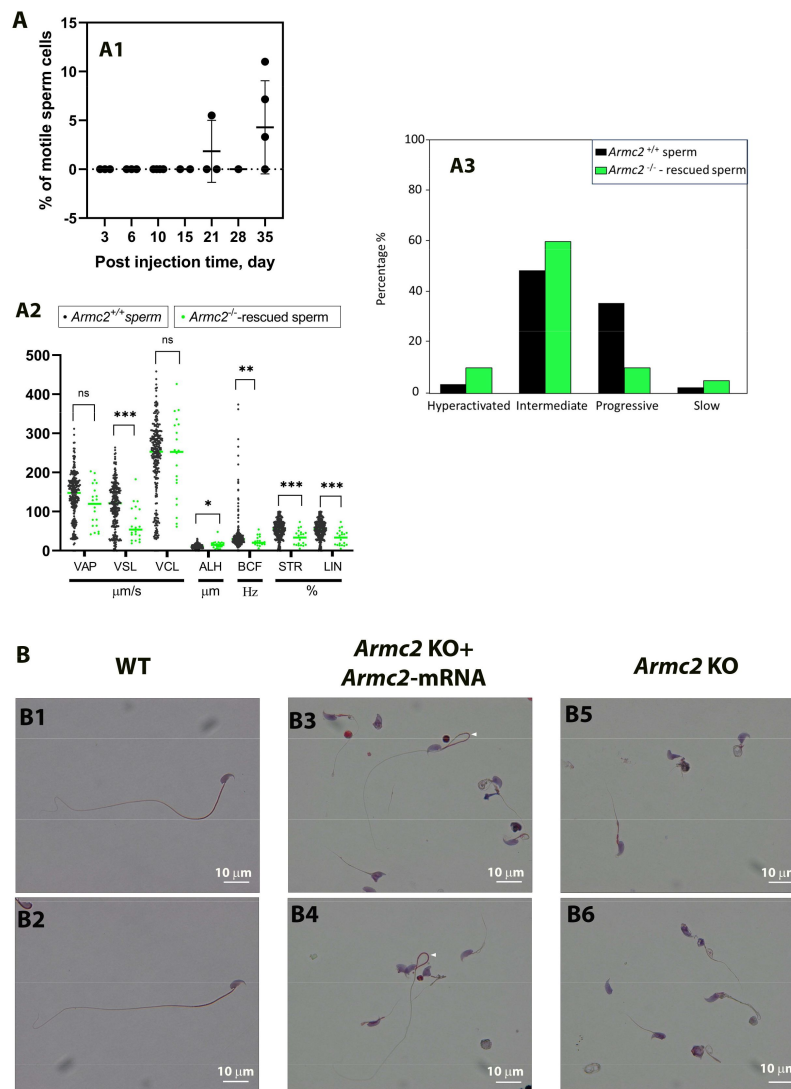


Figure 11

Sperm motility is restored in *Armc2* KO mice at 21 and 35 days after injection and electroporation of *Armc2*-mRNA.

(A) Adult *Armc2* KO mouse testes were injected with a solution containing *Armc2*-mRNA. After injection, the testes were electroporated. At different times (3-, 6-, 10-, 15-, 21-, 28-, and 35-days *post*-injection), sperm were extracted from the cauda epididymis of the injected testis, and the sample was then examined with a CASA system to identify the percentage of motile spermatozoa (A1). $n = 2$ for day 15, $n = 3$ for days 3, 6 and 21, $n = 4$ for day 10, and $n = 5$ for days 28 and 35. (A2) Sperm motility parameters of *Armc2*^{-/-}-rescued sperm in comparison to *Armc2*^{+/+} sperm. The motility parameters measured were: averaged path velocity (VAP); straight line velocity (VSL); curvilinear velocity (VCL); amplitude of lateral head displacement (ALH); beat cross frequency (BCF); straightness (STR); linearity (LIN). Black dots: sperm cells from *Armc2* null mice, green dots: sperm cells from *Armc2* null mice 35 days after injection with *Armc2*-mRNA. Results are expressed as mean $\pm$ SD. (A3) Sperm motility population of *Armc2*^{-/-}-rescued sperm in comparison to *Armc2*^{-/-} sperm. Black column: sperm cells from *Armc2* null mice, green column: sperm cells from *Armc2* null mice 35 days after injection with *Armc2*-mRNA. Statistical significance was verified using a Mann-Whitney sum test. Data are displayed as mean $\pm$ SEM. P values of $* \leq 0.05$, $** \leq 0.01$, or $*** \leq 0.001$ were considered to represent statistically significant differences.

(B) Morphology of sperm cells in *Armc2* KO mice injected or not with *Armc2*-mRNA. (B1-B2): microscopic observation of epididymal sperm cells from a mature WT mouse. (B3-B4): epididymal sperm cells from a mature *Armc2* KO mouse 35 days after injection/electroporation with *Armc2*-mRNA. (B5-B6): epididymal sperm cells from a control *Armc2* KO male. Normal sperm cells were observed in the injected condition with *Armc2*-mRNA. (White arrows), Scale bars: 10 μ m.

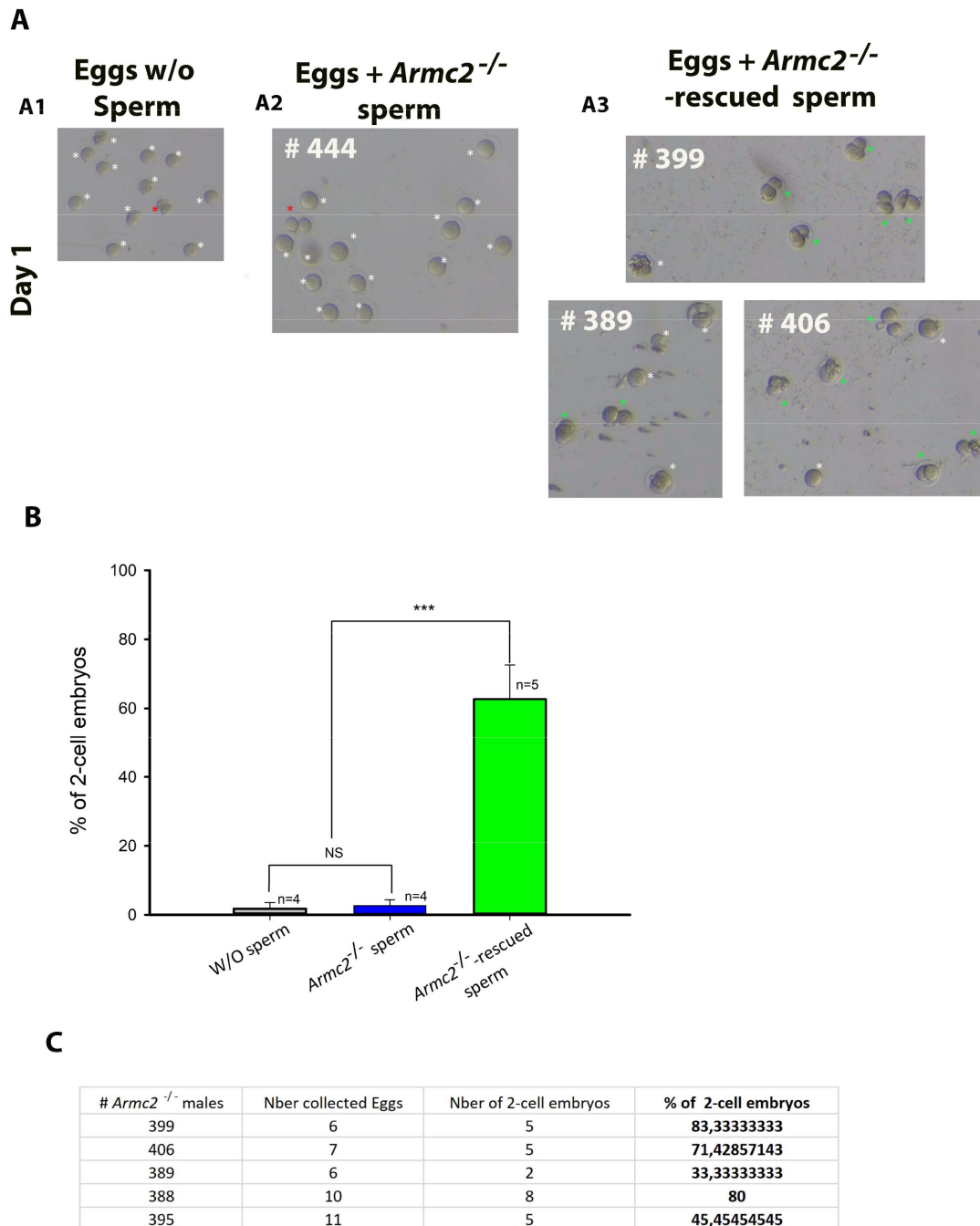


Fig 12

***Armc2*^{-/-}-rescued sperm can fertilize eggs and produce embryos by IVF.**

(A) Illustrations showing eggs/embryos obtained 24 h after egg collection in the following conditions. (A1) W/O sperm (A2) IVF with *Armc2*^{-/-} sperm from *Armc2*^{-/-} males (#444) and (A3) IVF with *Armc2*^{-/-} rescued sperm from *Armc2*^{-/-} males treated with mRNA-*Armc2* (males #399, #389, and #406). Green and red asterisks show 2-cell embryos, red asterisks showing 2-cell embryos obtained by parthenogenesis. White asterisks show unfertilized oocytes or degenerated. (B) Histograms showing the mean percentage $\pm$ SD of alive eggs reaching the 2- cell embryo stage at 24 hours after IVF without sperm (n=4), with *Armc2*^{-/-} sperm (n=4), and with *Armc2*^{-/-} rescued sperm (n=5). Statistical significance was verified using a one-way ANOVA test. (C) The table presents the data pertaining to the number of two-cell embryos obtained by IVF with *Armc2*^{-/-} rescued sperm from five different *Armc2*^{-/-} males treated with mRNA- *Armc2* (males #399, #406, #389, #388, and #395). The data is presented in terms of the percentage of two-cell embryos obtained in relation to the total number of eggs collected.

rescued spermatozoa and 13% for *Armc2*^{-/-} spermatozoa (Supp Fig.9 [↗](#)). The findings indicate that the developmental potential of the embryos was enhanced when *Armc2*^{-/-}-rescued sperm were utilized as opposed to *Armc2*^{-/-} sperm.

Overall, these results demonstrate that the *Armc2*^{-/-}-rescue motile spermatozoa can successfully fertilize eggs and produce embryos capable of developing properly into blastocysts.

Discussion

The challenge of treating male infertility remains to be addressed. Current assisted reproduction techniques (ARTs) are unable to treat all patients, and alternative strategies need to be developed to meet the legitimate desire to be a father. The aim of this study was to evaluate the potential of naked mRNA as a means to induce an expression of exogenous proteins in male germ cells in a preclinical adult mouse model. Based on previous studies using electroporation, we investigated whether the combination of the injection of naked mRNA and *in vivo* electroporation could lead to an efficient protein expression in spermatogenic cells. We chose to first study the efficiency of capped and poly(A)-tailed mRNA coding for reporter proteins and compared the results to those obtained with a non-integrative enhanced episomal vector plasmid. No EEV plasmid has ever been tested in the context of infertility treatment before this study.

Using an adult mouse model, we optimized the micro-injection and electroporation method described by Michaelis et al [[15](#) [↗](#)]. We show that the microinjection through the *rete testis* did not provide a homogenous distribution of the particles throughout the seminiferous tubules. Nevertheless, the seminiferous tubules remained intact, with no signal detected in the peritubular space. The peripheral expression observed was due to the close vicinity of cells to the electrodes, and to a peripheral dispersal of the injected solution, as shown by the distribution of the fluorescent i-particles NIRFiP-180. Our results also showed that the combination of injection and electroporation did not perturb spermatogenesis when electric pulses are carefully controlled. Using such a protocol, we were able to induce the expression of 3 reporter proteins, GFP, mCherry and luciferase in the testis by mRNA injection/electroporation.

Using whole testicular optical clearing, the reporter proteins were synthesized from the injected mRNA in different types of cells including Sertoli cells, spermatogonia, spermatocytes, round spermatids, and mature spermatids from day 1, and were still detectable after 1 week. These results deserve two comments: first, the expression is very fast and synthesized protein is detectable 24 h injection, second all cell types have the ability to translate mRNAs. Furthermore, the fact that we observed motile sperm at 21 days after injection confirms that spermatids are transfected and that the translation of the protein of interest is possible at this stage. For EEV, we have a similar result at day 1. However, the yields of seminiferous tubules and cellular transfection are lower. In particular, a lower level of transfection of germ cells was observed than with the mRNA. It is worth noting that after one week, the reporter proteins synthesized from injected EEV, were only discernible in the Sertoli cells.

Based on whole testes fluorescence and, for the first time, *in vivo* bioluminescence imaging of testes, we characterized the kinetics of mRNA expression. The signal measured is the fluorescence or the bioluminescence emitted by the GFP or luciferase. This signal is dependent on both the half-lives of the plasmid/mRNA and the proteins. Therefore, the kinetic of the signal persistence is a combination of the persistence of the vector and the synthesized protein. For mRNA, it's difficult to determine the lifespan of our mRNAs because these mRNAs have been modified at different levels including 5'CAP, mRNA body, poly(A)tail, which increase mRNA stability and translation [[30](#) [↗](#)]. Nevertheless, their half-lives should not exceed a few days and therefore the fluorescent signal observed, ranging from 15- to 21-days, depending on the molecule being expressed, likely corresponds to the persistence of the protein synthesized during the time window of mRNA

expression. The persistence of the reporter proteins is in line with the fact that proteins involved in spermatogenesis exhibit a markedly low turnover rate [31]. This is due to the fact that these proteins are stored within sperm organelles, such as the acrosome, manchette, centrioles or fibrous sheath. These organelles, made during spermiogenesis, remain stable for weeks until the fertilization process occurs because there is no protein synthesis in mature sperm. For example, the Cav3.2 calcium channel is expressed during meiosis at the pachytene stage and contributes to calcium signaling during acrosome reaction [32–35]. When using the EEV vector, expression persisted for longer – up to 119 days – due to the intrinsic property of the EEV plasmid which allows its replication in synchronous manner with the host genome.

These results suggest that although EEV expression lasted longer, mRNAs, by targeting more efficiently male germ cells and allowing higher transfection yields of seminiferous tubules, could be a more effective and potent tool to express exogenous proteins in germ cells. By expressing a missing protein in the case of male infertility due to monogenic causes, it could be possible to restore failed spermatogenesis and thus to treat infertility.

ARMC2 is expressed in late spermatogenesis stages

We show that ARMC2 was localized in the flagellum of spermatids obtained by enzymatic dissociation. No *Armc2* expression was detected in earlier germ cell type lineages like spermatogonia or spermatocytes. These results suggest that the ARMC2 protein is expressed late during spermatogenesis, which explains why motile sperm were found in the cauda epididymis from 3-weeks after injection in our treated mice. Indeed, full spermiogenesis (from round cells to sperm) takes around 15 days [36], and the journey across the epididymis lasts around 8 days, making a total of 3-weeks. Our results also confirm those recently published by Lehtreck et al. [37] from their study of the role of ARMC2 in the Intra-Flagellar Transport (IFT) of radial spokes in *Chlamydomonas*. They suggested that the transport of the radial spokes along the flagellum involves ARMC2, acting as an IFT adapter [37]. The presence of ARMC2 in the flagella of elongating spermatid supports this hypothesis.

Exogenous *Armc2*-mRNA expression rescued the motility of oligo-astheno- teratozoospermic sperm

This is the first demonstration that proteins can be expressed in the testis following electroporation with optimized, capped and poly(A)-tailed mRNA.

Our objective was to develop a new targeted therapeutic approach for infertility associated with monogenic defects. The objective of this preclinical study was to ascertain the efficacy of messenger RNA (mRNA) in expressing a missing protein, ARMC2, in a mouse model exhibiting oligo-astheno-teratozoospermia due to the absence of *Armc2*, with the aim of restoring flagellar motility and fertility. Our results strongly suggest that the strategy did not alter spermatogenesis, as injection and electroporation of *Armc2*-mRNA or EEV-*Armc2* had no effect on testicular morphology or weight. More importantly, the technique was effective, with motile sperm cells found in cauda epididymis 3 and 5 weeks after *Armc2*-mRNA injection into testes from *Armc2* KO males. Nevertheless, it should be noted that not all injected mice were efficiently treated. For example, only 87.5% of the treated mice (14 of the 16) exhibited motile sperm after 5 weeks. The absence of motile sperm at 5 weeks may be attributed to the specific types of spermatogenic cells that were transfected during the electroporation phase. It is possible that the transfected cells may differ between individuals, potentially influenced by the injection and the position of the electrodes during electroporation. If the mRNA transfection occurs in a spermatogonia, it may take more than six weeks (including the time required to cross the epididymis) before motile epididymal sperm cells emerge. This potential timeline could explain the absence of motile sperm in some mice at 5 weeks.

Due to the quantity of motile sperm obtained, it was not possible to produce offspring through natural mating. However, the *Armc2*^{-/-}-rescued sperm exhibited normal morphology, motility, and *in vitro* fertility. Indeed, the *Armc2*^{-/-}-rescued motile spermatozoa have successfully fertilized eggs and produced embryos that were capable of developing properly into blastocysts. These results provide compelling evidence that the method can effectively produce fertile sperm. It worth to note that the significant modifications of the CASA parameters observed for rescued sperm did not impact their fertilizing potential. Naked mRNA injection/electroporation is therefore a promising method to treat infertility. In contrast, no motile spermatozoa were found after injection/electroporation of EEV-*Armc2*, confirming our previous results suggesting that this nucleic tool does not efficiently enter or transfect germ cells.

Despite this success, the transfection rate deserves to be improved to obtain a larger number of sperm cells to produced offspring through natural mating. To increase the testicular transfection rate, encapsulation of mRNA into lipids nanoparticles could be used, as used for Covid vaccination [38]. During the writing of this manuscript, Dong team [39], used a self- amplifying RNA (saRNA) encapsulated in cholesterol-amino-phosphate derived lipid nanoparticle to restore spermatogenesis in infertile mice. They successfully restore the expression of the DNA Meiotic Recombinase 1 (DMC1) [40–42] in *Dmc1* KO infertile mice by injecting a self-amplifying RNA-*Dmc1* in the testes. saRNA are genetically engineered replicons derived from self-replicating single-stranded RNA viruses [43]. The saRNA contains the alphavirus replicase genes and encodes an RNA-dependent RNA polymerase (RdRP) complex which amplifies synthetic transcripts *in situ* and the target RNA sequence. The target RNA is expressed at high levels as a separate entity. As a result of their self-replicative activity, saRNAs can be delivered at lower concentrations than conventional mRNA to achieve comparable expression levels [44]. Moreover, saRNA constructs need to be condensed by a cationic carrier into a nanoparticle measuring ~100 nm to enable their uptake into target cells and protect the saRNA from degradation [45]. Finally, saRNA will amplify the RNA without cellular regulation. For all these reasons, if such a strategy is to be pursued, a potential toxicity effect due to saRNA over expression must be investigated in the testes and progeny. Another difficulty with saRNA relates to the size of the molecular construct. saRNA sequences are large and complex. The length of the sequence RdRP is around 7 kilobases, which often makes the full length of saRNA more than 9 kilobases once the sequence for the protein of interest has been integrated [45]. Dong et al. [39] successfully used their saRNA construct to rescue spermatogenesis failure induced by the absence of the small protein *Dmc1* (37 KDa), but it may be more challenging with larger proteins such as the structural proteins involved in OAT, including the 98-kDa ARMC2 [23].

Our next step will be to assess whether encapsulating *Armc2*-mRNA in LNP-CAP could allow a larger number of germ cells to be transfected.

Naked mRNA, a new therapeutic strategy to treat severe infertility

Non-obstructive Azoospermia (NOA) and severe oligozoospermia (SO) are the most severe disorders of spermatogenesis and are the most likely to be of genetic origin. NOA is defined by the complete absence of spermatozoa in the ejaculate. Approximately 10–15 % of infertile men have azoospermia, and a further 15 % have SO [46]. For patients with NOA, few clinical solutions are currently available. Generally, testicular sperm extraction is attempted to collect some spermatozoa from the seminiferous tubules, which can then be used for ICSI [47]. When no sperm are retrieved, intra-conjugal conception is impossible. The results of this study strongly suggest that transient mRNA expression of a missing protein in NOA testes by electroporation could be sufficient to produce normal sperm for IVF and obtain embryos.

In conclusion, this paper presents the first *in vivo* testicular injection and electroporation of capped and poly(A)-tailed mRNA, demonstrating that it is an efficient strategy to transfect male germ cells and that the duration of the expression of the resulting proteins is compatible with restoring spermatogenesis. Our comprehensive study revealed mRNA to be more efficient than an

episomal vector, despite longer-lived expression in male germ cells with EEV. The difference was linked to EEV achieving a lower rate of seminiferous tubule transfection and a shorter duration of expression in germ cells. Our findings have also demonstrated, for the first time, that sperm motility and fertility can be restored in mice with an oligo-astheno-teratozoospermia phenotype through a technique that combines injection and electroporation of capped and poly(A)-tailed mRNA. The findings presented open new opportunities to develop efficient strategies to treat male infertility with monogenic causes.

Acknowledgements

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Additional information

Ethics statement

All procedures involving animals were performed in line with the French guidelines for the use of live animals in scientific investigations. The study protocol was approved by the local ethics committee (ComEth Grenoble # 318) and received governmental authorization (ministerial agreement # 38109-2022072716142778).

Additional files

Video 1. D-microscopic reconstructions of face 1 of a testis injected with GFP-mRNA. [↗](#)

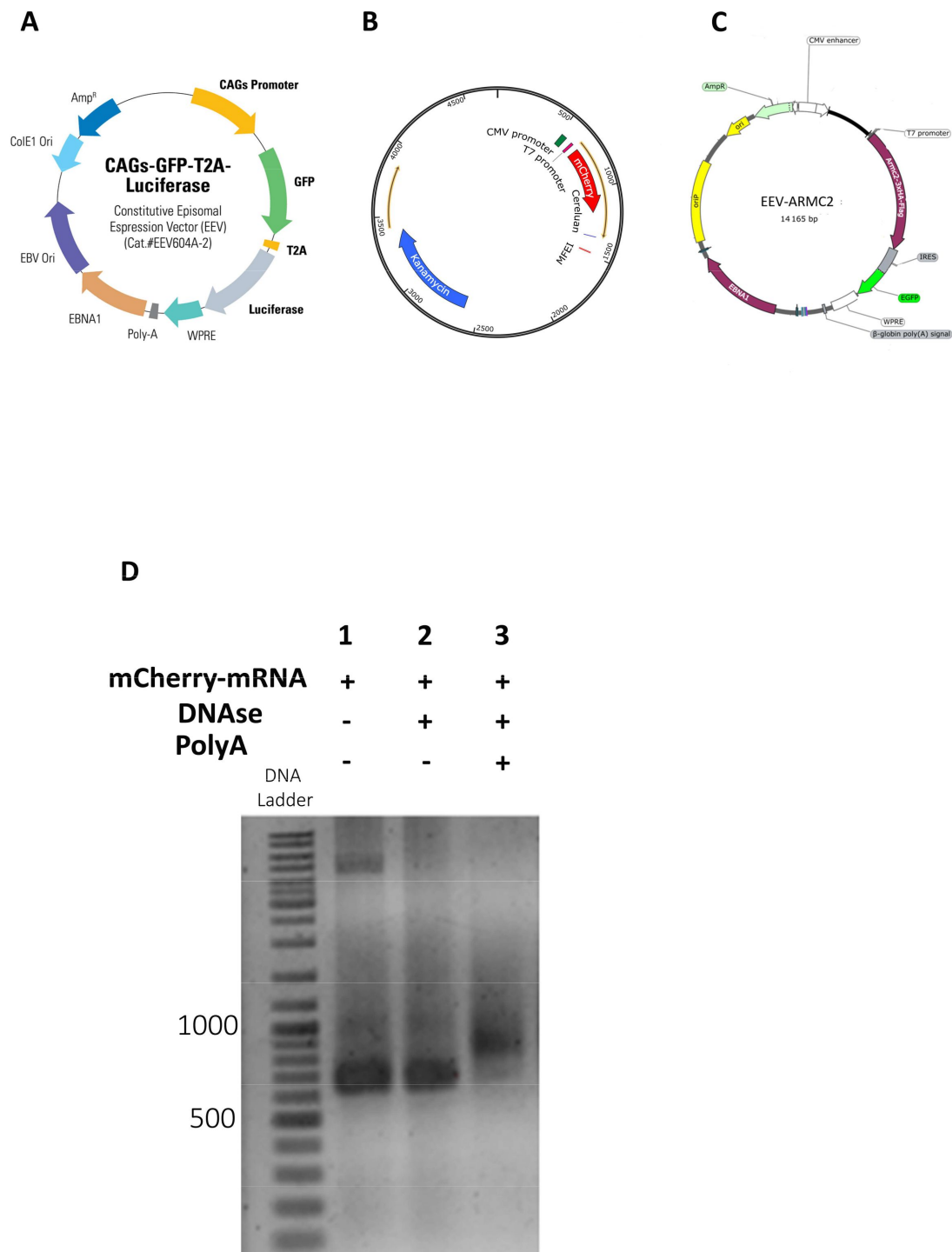
Video 2. D-microscopic reconstructions of face 2 of a testis injected with GFP-mRNA. [↗](#)

Video 3. CASA recording of WT epididymal sperm cells. [↗](#)

Video 4. CASA recording of *Armc2* KO epididymal sperm cells. [↗](#)

Video 5. CASA recordings of epididymal sperm cells from *Armc2* KO mice on day 21 after injection/electroporation with *Armc2*-mRNA. [↗](#)

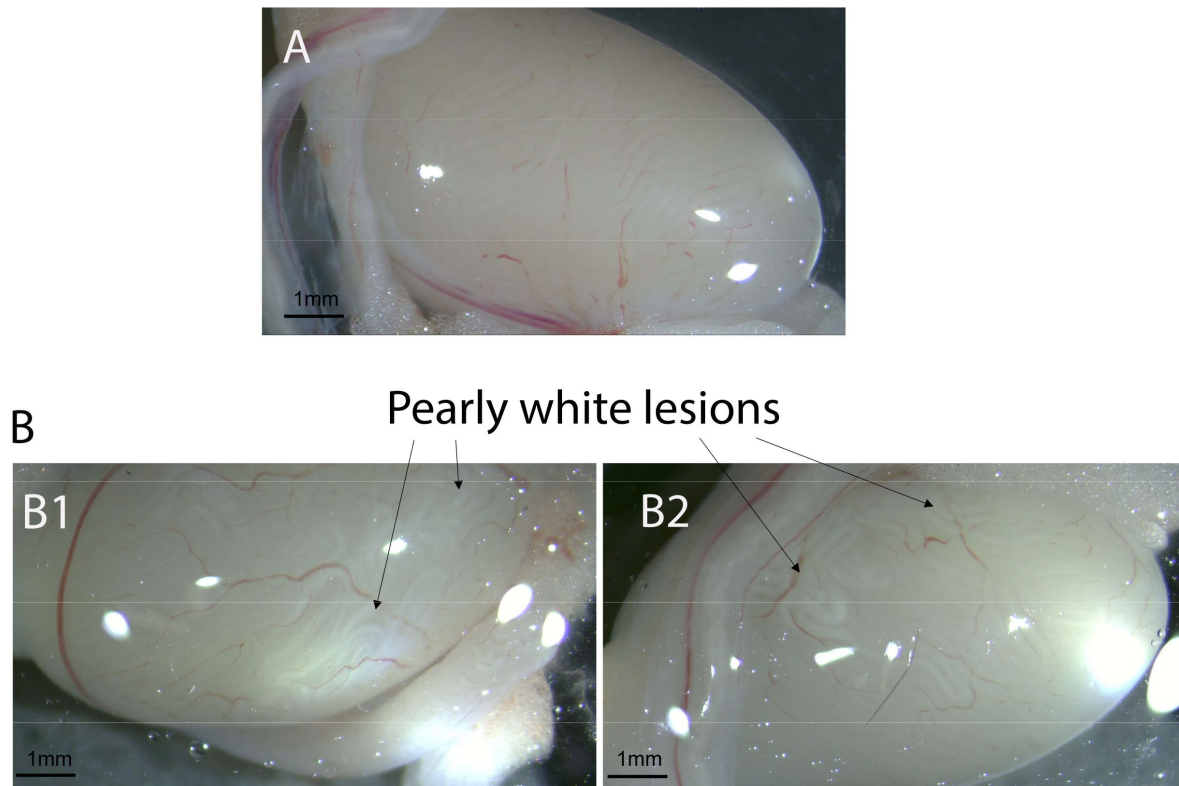
Video 6. CASA recordings of epididymal sperm cells from *Armc2* KO mice on day 21 after injection/electroporation with *Armc2*-mRNA. [↗](#)



Supp Fig 1

EEV and mRNA maps

(A) EEV-plasmid map. The EEV-plasmid contains GFP, EVB ori, GFP, Luciferase, and EBNA sequences under the control of the GAGs promoter. (B) The mCherry plasmid contains the mCherry gene under the control of a T7 promoter. (C) The EEV-*Armc2* plasmid contains GFP, oriP, EBNA and *Armc2* sequences under the control of the CMV and T7 promoters. (D) *mCherry*-mRNA was synthesized as described in Material and Methods. It was validated by agarose gel electrophoresis: Lane 1: DNA size marker ladder (100 bp), lane 2: capped *mCherry*-mRNA (IVT product), lane 3: *mCherry*-mRNA after DNase treatment. Capped and poly A tailed *mCherry*-mRNA migrated to the expected size of 876 bp.

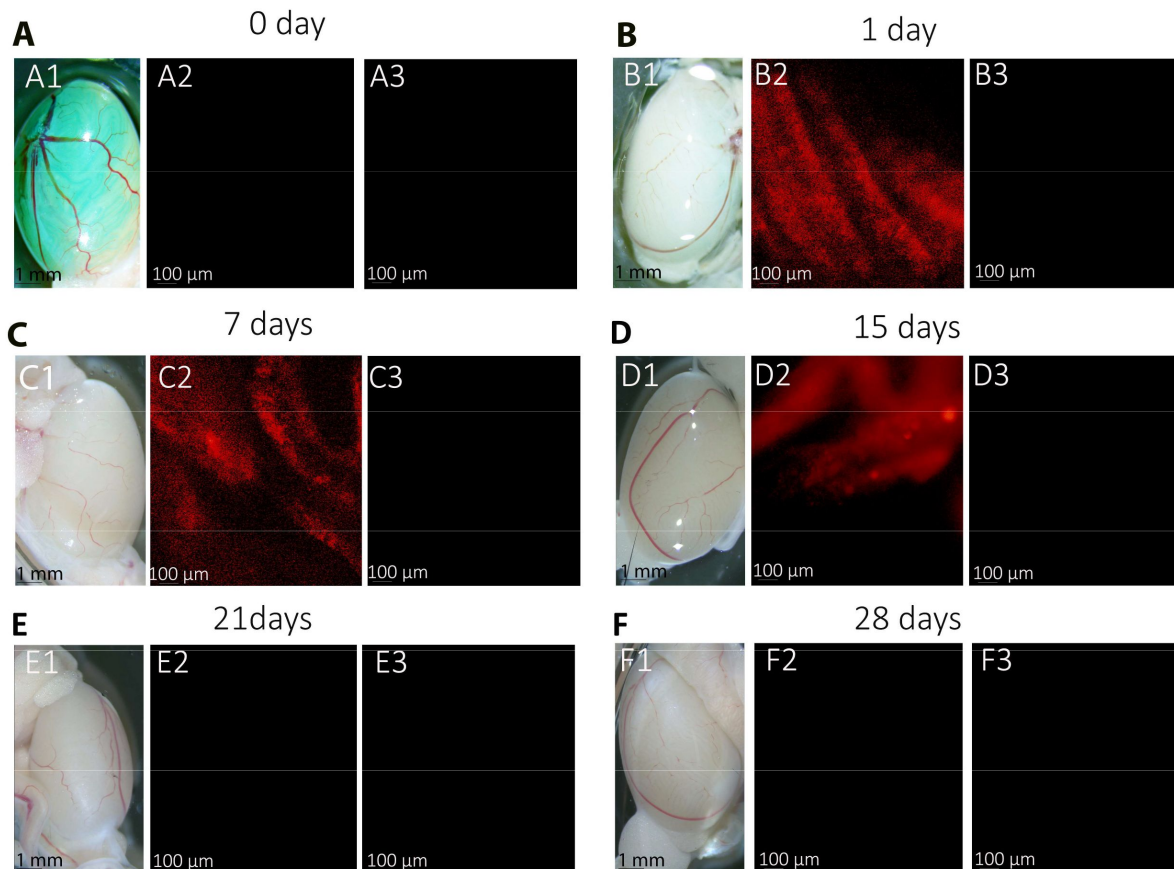


Supp Fig 2

Damaged tubules observed by optical microscopy following overstimulation

Adult mouse testes were *in vivo* injected and over electroporated using 10 square electric pulses to induce damage. (A) Control testis (no injection/electroporation). (B1-B2) Over electroporated testes showing damaged tubules as pearly white striations. Scale bars: 1 mm.

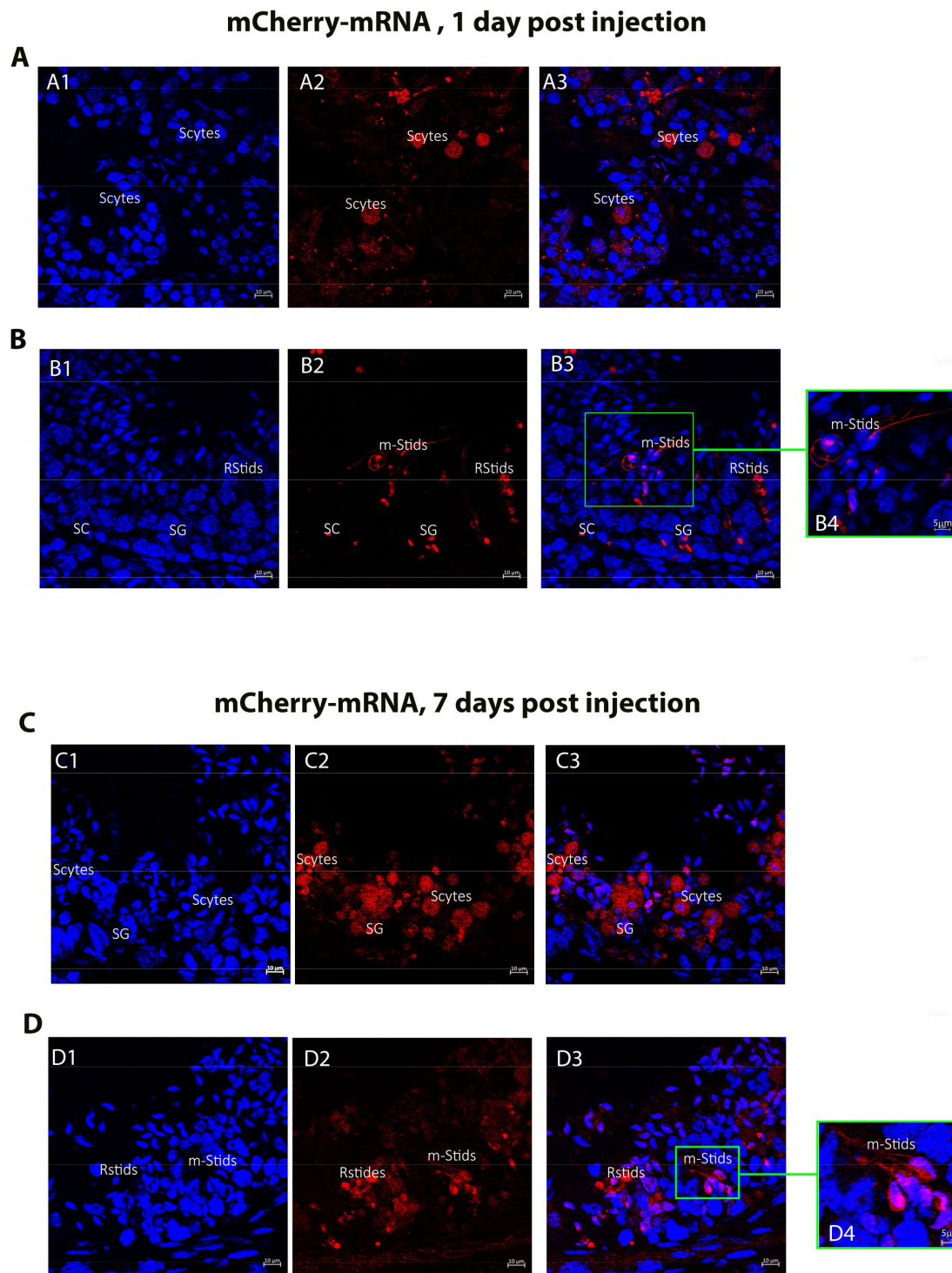
mCherry-mRNA



Supp Fig 3

Testicular expression of *mCherry*-mRNA following *in vivo* electroporation

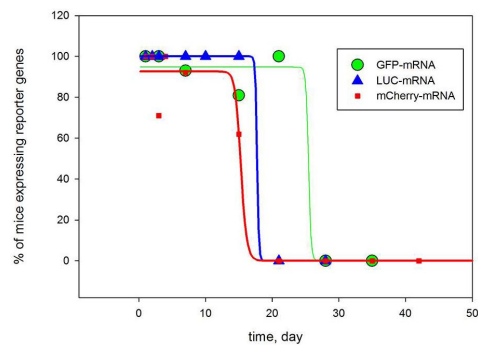
(A1, B1, C1, D1, E1 and F1) Whole-mount testes on days 0, 1, 7, 14, 21, 28 after *in vivo* injection/ electroporation. (A2, B2, C2, D2, E2 and, F2) Using fluorescence microscopy, transfected testes from 12-week-old B6D2 mice express red mCherry fluorescence. mCherry was detected in a diffuse pattern throughout the seminiferous tubules from day 1 to day 15. (A3, B3, C3, D3, E3 and F3) images showing the absence of autofluorescence in non- transfected control testes observed at 4x magnification. Scales bars: 1 mm and 100 μm.



Supp Fig 4

Cellular expression of *mCherry*-mRNA following in vivo injection/ electroporation.

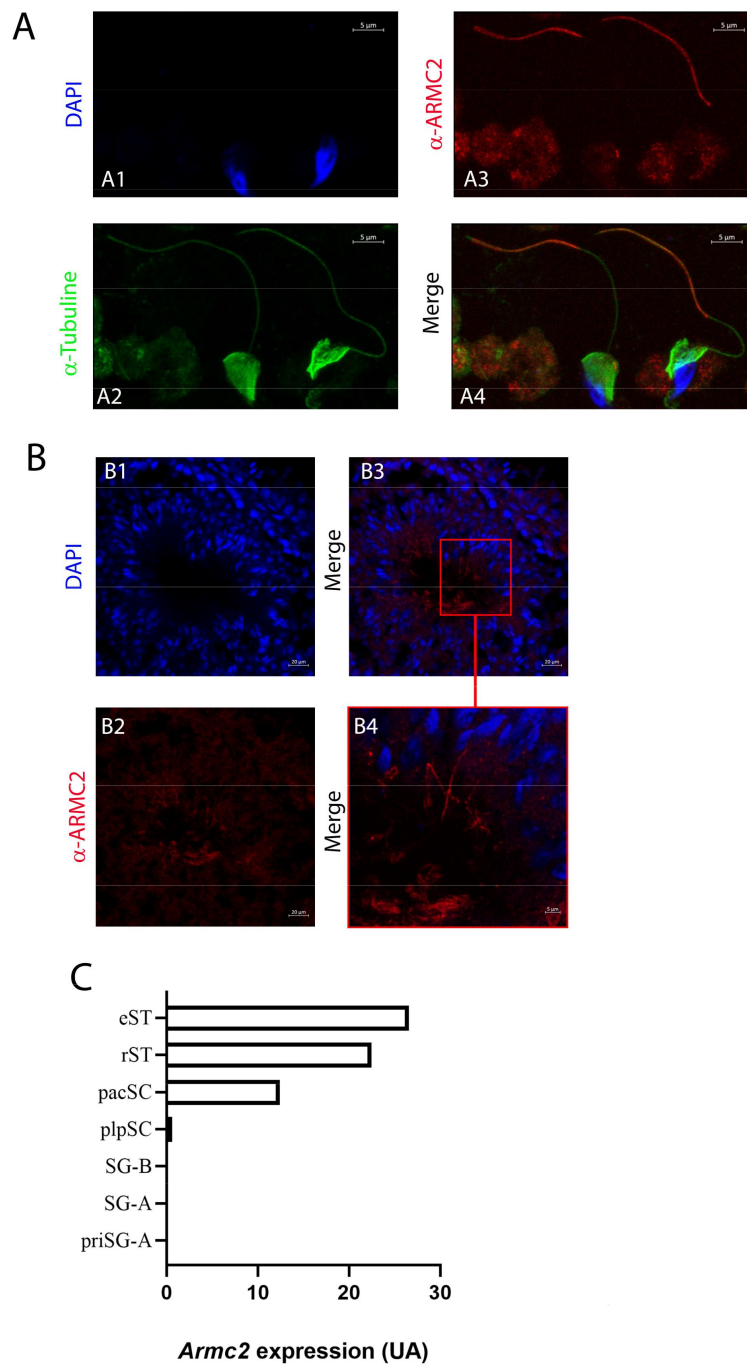
Cross sections (20 μm) of mouse testes on day 1 (AB) and day 7 (CD) after in vivo injection and electroporation with *mCherry*-mRNA, observed under fluorescence microscopy. Red signals correspond to successfully transfected testicular tubular cells; nuclei were counterstained with DAPI (blue). At the cellular level, mCherry fluorescence was detectable in Sertoli cells (SC); Spermatogonia (SG); Spermatocytes (Scytes); round Spermatids (Rstids), and mature spermatids (m-Sptids); Scale bars: 10 μm and 5 μm .



Supp Fig 5

Decay over time of the number of mice exhibiting reporter gene expression following injection/electroporation of the three different mRNAs.

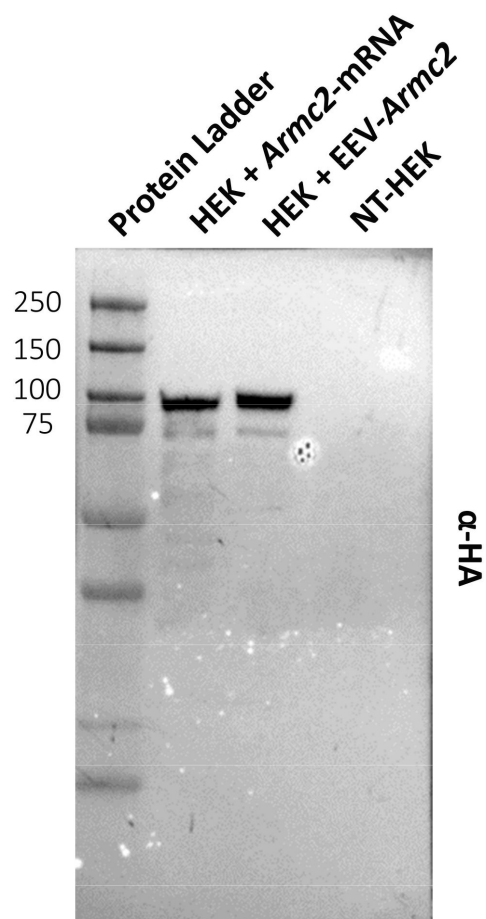
Mice were injected on day 0 with *LUC*-mRNA, *GFP*-mRNA or *mCherry*-mRNA and the number of mice showing bioluminescence or fluorescence in the testis was counted at different time points. For *LUC*-mRNA n= 5 mice at each time point. For *mCherry*-mRNA n=3 on day 1; n=4 on day3; n=15 on day 7; n= 21 on day 15; n=15 on day 21; n= 5 on day 28; n=5 on day 35; and for *GFP*-mRNA n=3 on day 12; n=7 on day 2; n=7 on day 3; n=12 on day 7; n= 13 on day 15; n=10 on day 21 ; n= 9 on day 28; n=17 on day 35 and n=5 on day 42.



Supp Fig 6

Armc2 expression and localization in mice testis

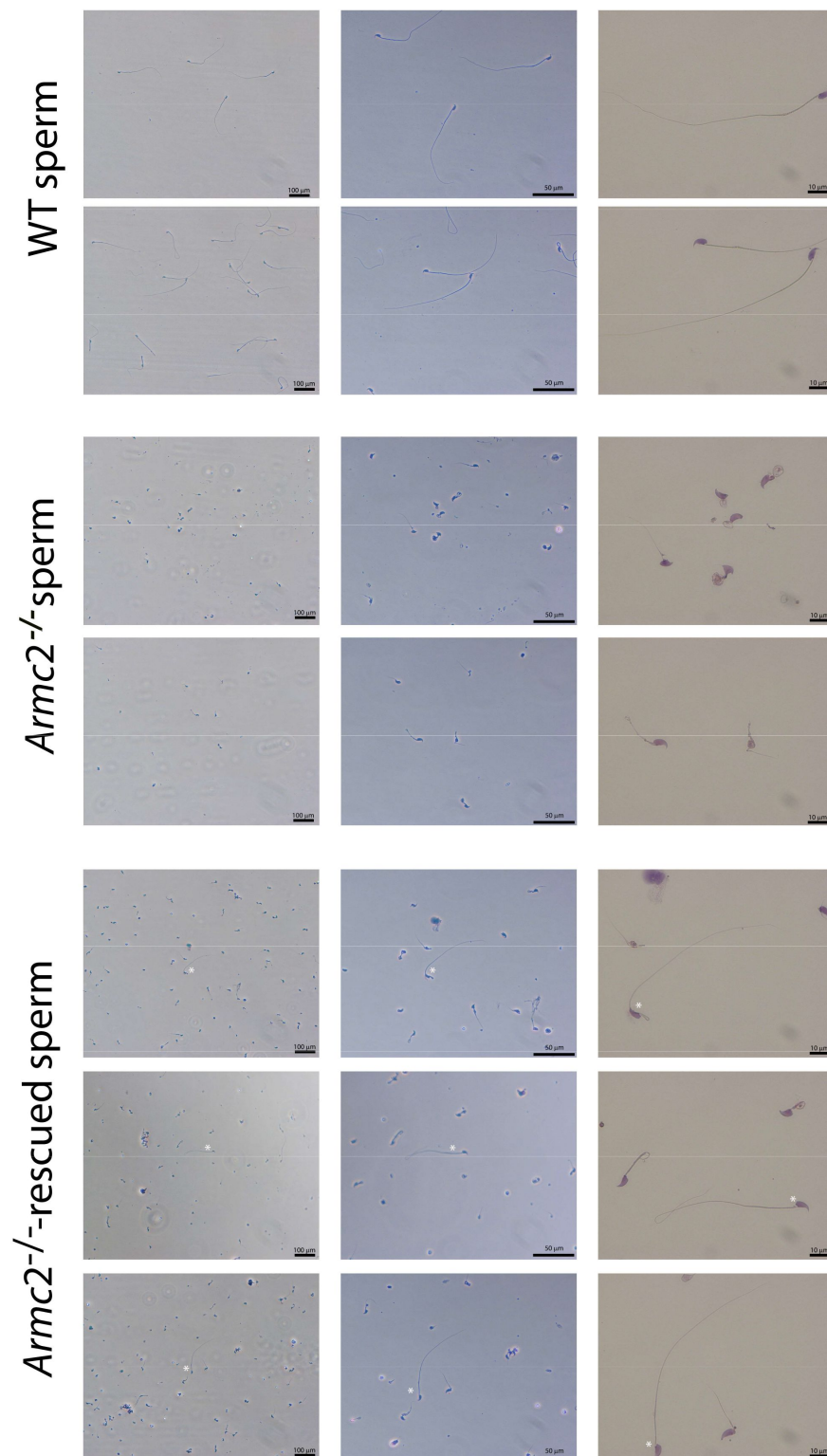
(A) IF experiment on dissociated spermatogenic cell from WT male. (A1) Nuclei were counterstained with DAPI (blue staining), (A2) tubulin (green signal) and (A3) ARMC2 (red signal). (A4) overlay. ARMC2 is located in the flagellum of spermatids Scale bars: 5 μ m. (B) Cross-sections of seminiferous tubules (B1) Nuclei were counterstained with DAPI (blue staining), (B2) and (B3) ARMC2 (red signal). (B4) is an enlargement of the red square in B3, showing that only elongating/mature spermatids were stained. (C) Expression of *Armc2* in mouse spermatogenic cells based on the RNA-sequencing study from Gan et al. 2013. Primitive type A spermatogonia (priSG-A) were isolated from 6 dpp mice; Type A spermatogonia (SG-A) and type B spermatogonia (SG-B) were from 8 dpp mice; preleptotene spermatocytes (plpSC) and pachytene spermatocytes (pacSC) were from 17 dpp mice; and round spermatids (rST), elongating spermatids (eST) and spermatozoa (SZ) were from 60 dpp mice. Artificial units (AU). Dpp: day post-partum



Supp Fig 7

Validation of *Armc2*-mRNA in HEK cells.

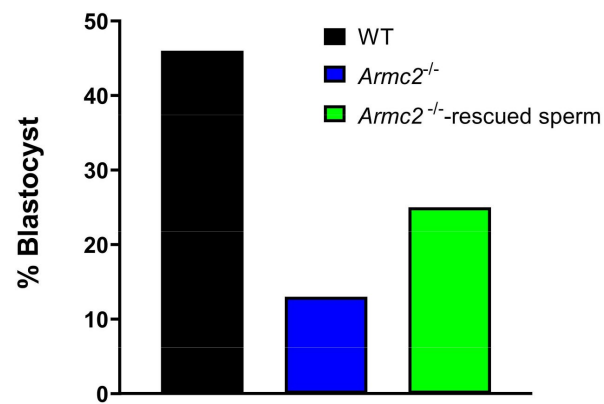
HEK cells were transfected with *Armc2*-mRNA or EEV-*Armc2*. ARMC2 protein was detected by Western blot with an anti-HA primary antibody. The expected size of the ARMC2 protein is 98 kDa.



Supp Fig 8

Morphology of sperm cells from WT, *Armc2* KO and *Armc2*-rescued males.

Microscopic observation of epididymal sperm cells from WT, *Armc2* KO and *Armc2* KO-*Armc2*- mRNA injected males at 20x (left column), 40x (middle column) and 100x (right column) magnifications. In the rescue condition, rescued sperm cells were labeled with a white asterisk. Scale bars: 100, 50 and 10 μm.



Supp Fig 9

***Armc2*^{-/-}-rescued sperm can fertilize eggs and produce embryos by ICSI.**

Comparative analysis of the percentage of blastocysts produced by ICSI with spermatozoa from wild-type (WT), *Armc2*^{-/-} and *Armc2*^{-/-}-rescued individuals. For *Armc2*^{-/-}-rescued individuals, only motile sperm were injected.

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Reviewer #1 (Public review):

The authors assess the effectiveness of electroporating mRNA into male germ cells to rescue the expression of proteins required for spermatogenesis progression in individuals where these proteins are mutated or depleted. To set up the methodology, they first evaluated the expression of reporter proteins in wild-type mice, which showed expression in germ cells for over two weeks. Then, they attempted to recover fertility in a model of late spermatogenesis arrest that produces immotile sperm. By electroporating the mutated protein, the authors recovered the motility of ~5% of the sperm; although the sperm regenerated was not able to produce offspring using IVF, the embryos reached the 2-cell state (in contrast to controls that did not progress past the zygote state).

This is a comprehensive evaluation of the mRNA methodology with multiple strengths. First, the authors show that naked synthetic RNA, purchased from a commercial source or generated in the laboratory with simple methods, is enough to express exogenous proteins in testicular germ cells. The authors compared RNA to DNA electroporation and found that germ cells are efficiently electroporated with RNA, but not DNA. The differences between these constructs were evaluated using in vivo imaging to track the reporter signal in individual animals through time. To understand how the reporter proteins affect the results of the experiments, the authors used different reporters: two fluorescent (eGFP and mCherry) and one bioluminescent (Luciferase). Although they observed differences among reporters, in every case expression lasted for at least two weeks.

The authors used a relevant system to study the therapeutic potential of RNA electroporation. The ARMC2-deficient animals have impaired sperm motility phenotype that affects only the later stages of spermatogenesis. The authors showed that sperm motility was recovered to ~5%, which is remarkable due to the small fraction of germ cells electroporated with RNA with the current protocol. The sperm motility parameters were thoroughly assessed by CASA. The 3D reconstruction of an electroporated testis using state-of-the-art methods to show the electroporated regions is compelling.

The main weakness of the manuscript is that although the authors manage to recover motility in a small fraction of the sperm population, it is unclear whether the increased sperm quality is substantial to improve assisted reproduction outcomes. The authors found that the rescued sperm could be used to obtain 2-cell embryos via IVF, but no evidence for more advanced stages of embryo differentiation was provided. The motile rescued sperm

was also successfully used to generate blastocyst by ICSI, but the statistical significance of the rate of blastocyst production compared to non-rescued sperm remains unclear. The title is thus an overstatement since fertility was never restored for IVF, and the mutant sperm was already able to produce blastocysts without the electroporation intervention.

Overall, the authors clearly show that electroporating mRNA can improve spermatogenesis as demonstrated by the generation of motile sperm in the ARMC2 KO mouse model.

<https://doi.org/10.7554/eLife.94514.2.sa3>

Reviewer #2 (Public review):

The authors inject, into the rete testes, mRNA and plasmids encoding mRNAs for GFP and then ARMC2 (into infertile *Armc2* KO mice) in a gene therapy approach to express exogenous proteins in male germ cells. They do show GFP epifluorescence and ARMC2 protein in KO tissues, although the evidence presented is weak. Overall, the data do not necessarily make sense given the biology of spermatogenesis and more rigorous testing of this model is required to fully support the conclusions, that gene therapy can be used to rescue male infertility.

In this revision, the authors attempt to respond to the critiques from the first round of reviews. While they did address many of the minor concerns, there are still a number to be addressed. With that said, the data still do not support the conclusions of the manuscript.

(1) The authors have not satisfactorily provided an explanation for how a naked mRNA can persist and direct expression of GFP or luciferase for ~3 weeks. The most stable mRNAs in mammalian cells have half-lives of ~24-60 hours. The stability of the injected mRNAs should be evaluated and reported using cell lines. GFP protein's half-life is ~26 hours, and luciferase protein's half-life is ~2 hours.

(2) There is no convincing data shown in Figs. 1-8 that the GFP is even expressed in germ cells, which is obviously a prerequisite for the *Armc2* KO rescue experiment shown in the later figures! In fact, to this reviewer the GFP appears to be in Sertoli cell cytoplasm, which spans the epithelium and surrounds germ cells - thus, it can be oft-confused with germ cells. In addition, if it is in germ cells, then the authors should be able to show, on subsequent days, that it is present in clones of germ cells that are maturing. Due to intracellular bridges, a molecule like GFP has been shown to diffuse readily and rapidly (in a matter of minutes) between adjacent germ cells. To clarify, the authors must generate single cell suspensions and immunostain for GFP using any of a number of excellent commercially-available antibodies to verify it is present in germ cells. It should also be present in sperm, if it is indeed in the germline.

Other comments:

70-1 This is an incorrect interpretation of the findings from Ref 5 - that review stated there were ~2,000 testis-enriched genes, but that does not mean "the whole process involves around two thousand of genes"

74 would specify 'male'

79-84 Are the concerns with ICSI due to the procedure itself, or the fact that it's often used when there is likely to be a genetic issue with the male whose sperm was used? This should be clarified if possible using references from the literature, as this reviewer imagines this could be a rather contentious issue with clinicians who routinely use this procedure, even in cases where IVF would very likely have worked

199 Codon optimization improvement of mRNA stability needs a reference; in one study using yeast transcripts, optimization improved RNA stability on the order of minutes (e.g., from ~5 minutes to ~17 minutes); is there some evidence that it could be increased dramatically to days or weeks?

472-3 The reported half-life of EGFP is ~36 hours - so, if the mRNA is unstable (and not measured, but certainly could be estimated by qRT-PCR detection of the transcript on subsequent days after injection) and EGFP is comparatively more stable (but still hours), how does EGFP persist for 21 days after injection of naked mRNA??

Curious why the authors were unable to get anti-GFP to work in immunostaining?

In Fig. 3-4, the GFP signals are unremarkable, in that they cannot be fairly attributed to any structure or cell type - they just look like blobs; and why, in Fig. 4D-E, why does the GFP signal appear stronger at 21 days than 15 days? And why is it completely gone by 28 days? This data is unconvincing. If the authors did a single cell suspension, what types or percentage of cells would be GFP+? Since germ cells are not adherent in culture, a simple experiment could be done whereby a single cell suspension could be made, cultured for 4-6 hours, and non-adherent cells "shaken off" and imaged vs adherent cells. Cells could also be fixed and immunostained for GFP, which has worked in many other labs using anti-GFP.

In Fig. 5, what is the half-life of luciferase? From this reviewer's search of the literature, it appears to be ~2-3 h in mammalian cells. With this said, how do the authors envision detectable protein for up to 20 days from a naked mRNA? The stability of the injected mRNAs should be shown in a mammalian cell line - perhaps this mRNA has an incredibly long half-life, which might help explain these results. However, even the most stable endogenous mRNAs (e.g., globin) are ~24-60 hrs.

527-8 The Sertoli cell cytoplasm is not just present along the basement membrane as stated, but also projects all the way to the lumina

529-30 This is incorrect, as round spermatids are never "localized between the spermatocytes and elongated spermatids" - if elongated spermatids are present, rounds are not - they are never coincident in the same testis section

Fig. 7 To this reviewer, all of the GFP appears to be in Sertoli cell cytoplasm

In Figs 1-8 there is no convincing evidence presented that GFP is expressed in germ cells! In fact, it appears to be in Sertoli cells

Fig. 9 - alpha-tubuline?

Fig. 11 - how was sperm morphology/motility not rescued on "days 3, 6, 10, 15, or 28 after surgery", but it was in some at 21 and 35? How does this make sense, given the known kinetics of male germ cell development?? And at least one of the sperm in the KO in Fig. B5 looks relatively normal, and the flagellum may be out-of-focus in the image? With only a few sperm for reviewers to see, how can we know these represent the population?

<https://doi.org/10.7554/eLife.94514.2.sa2>

Reviewer #3 (Public review):

Summary:

The authors used a novel technique to treat male infertility. In a proof-of-concept study, the authors were able to rescue the phenotype of a knockout mouse model with immotile sperm

using this technique. This could also be a promising treatment option for infertile men.

Strengths:

In their proof-of-concept study, the authors were able to show that the novel technique rescues the infertility phenotype of *Armc2* knockout spermatozoa. In the revised version of the manuscript, the authors have added data on in vitro fertilisation experiments with *Armc2* mRNA-rescued sperm. The authors show that *Armc2* mRNA-rescued sperm can successfully fertilise oocytes that develop to the blastocyst stage. This adds another level of reliability to the data.

Weaknesses:

Some minor weaknesses identified in my previous report have already been fixed. The technique is new and may not yet be fully established for all issues. Nevertheless, the data presented in this manuscript opens the way for several approaches to immotile spermatozoa to ensure successful fertilisation of oocytes and subsequent appropriate embryo development.

[Editors' note: The images in Figure 12 do not support the authors' interpretation that 2-cell embryos resulted from in vitro fertilization. Instead, the cells shown appear to be fragmented, unfertilized eggs. Combined with the lack of further development, it seems highly unlikely that fertilization was successful.]

<https://doi.org/10.7554/eLife.94514.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

The authors assess the effectiveness of electroporating mRNA into male germ cells to rescue the expression of proteins required for spermatogenesis progression in individuals where these proteins are mutated or depleted. To set up the methodology, they first evaluated the expression of reporter proteins in wild-type mice, which showed expression in germ cells for over two weeks. Then, they attempted to recover fertility in a model of late spermatogenesis arrest that produces immotile sperm. By electroporating the mutated protein, the authors recovered the motility of ~5% of the sperm, although the sperm regenerated was not able to produce offspring using IVF.

We actually did not write that “sperm regenerated was not able to produce offspring using IVF” but rather that IVF was not attempted because the number of rescued sperm was too low. To address this important point, the ability of sperm to produce embryos was therefore challenged by two different assisted reproduction technologies, that are IVF and ICSI. To increase the number of motile sperm for IVF experiments, we have injected both testes from one male. We also conducted intracytoplasmic sperm injection (ICSI) experiments, using only rescued sperm, identified as motile sperm with a normal flagellum. The results of these new experiments have demonstrated that the rescued ARMC2 sperm successfully fertilized eggs and produced embryos at the two-cell stage by IVF and blastocysts by ICSI. These outcomes are presented in Figure 12.

This is a comprehensive evaluation of the mRNA methodology with multiple strengths. First, the authors show that naked synthetic RNA, purchased from a commercial source

or generated in the laboratory with simple methods, is enough to express exogenous proteins in testicular germ cells. The authors compared RNA to DNA electroporation and found that germ cells are efficiently electroporated with RNA, but not DNA. The differences between these constructs were evaluated using in vivo imaging to track the reporter signal in individual animals through time. To understand how the reporter proteins affect the results of the experiments, the authors used different reporters: two fluorescent (eGFP and mCherry) and one bioluminescent (Luciferase). Although they observed differences among reporters, in every case expression lasted for at least two weeks.

The authors used a relevant system to study the therapeutic potential of RNA electroporation. The ARMC2-deficient animals have impaired sperm motility phenotype that affects only the later stages of spermatogenesis. The authors showed that sperm motility was recovered to ~5%, which is remarkable due to the small fraction of germ cells electroporated with RNA with the current protocol. The 3D reconstruction of an electroporated testis using state-of-the-art methods to show the electroporated regions is compelling.

The main weakness of the manuscript is that although the authors manage to recover motility in a small fraction of the sperm population, it is unclear whether the increased sperm quality is substantial to improve assisted reproduction outcomes. The quality of the sperm was not systematically evaluated in the manuscript, with the endpoints being sperm morphology and sperm mobility.

We would like to thank the reviewers for their comments. As previously stated above, we produced additional rescue experiments and performed CASA, morphology observation, IVF and ICSI with the rescued sperm. The rescued ARMC2 sperm exhibited normal morphology (new figure 11 and Supp Fig 8), motility (figure 11), and fecundity (figure 12). Whereas sperm from untreated KO males were unable to fertilize egg by IVF, the rescued sperm fertilized eggs in vitro at a significant level (mean 62%, n=5), demonstrating that our strategy improves the sperm quality and assisted reproduction outcome (from 0 to 62%).

Some key results, such as the 3D reconstruction of the testis and the recovery of sperm motility, are qualitative given the low replicate numbers or the small magnitude of the effects. The presentation of the sperm motility data could have been clearer as well. For example, on day 21 after *Armc2*-mRNA electroporation, only one animal out of the three tested showed increased sperm motility. However, it is unclear from Figure 11A what the percentage of sperm motility for this animal is since the graph shows a value of >5% and the reported aggregate motility is 4.5%. It would have been helpful to show all individual data points in Figure 11A.

We provide now in figure 11A, a graph showing the percentage of rescued sperm for all animals. (scatter dot plot). Moreover, we performed additional CASA experiments to analyze in detail sperm motility (Figure 11A2-A3). Individual CASA parameters for motile sperm cells were extracted as requested by reviewer 3 and represented in a new graph (Fig 11 A2).

The expression of the reporter genes is unambiguous; however, better figures could have been presented to show cell type specificity. The DAPI staining is diffused, and it is challenging to understand where the basement membranes of the tubules are. For example, in Figures 7B3 and 7E3, the spermatogonia seems to be in the middle of the seminiferous tubule. The imaging was better for Figure 8. Suboptimal staining appears to lead to mislabeling of some germ cell populations. For example, in Supplementary Figure 4A3, the round spermatid label appears to be labeling spermatocytes. Also, in some instances, the authors seem to be confusing, elongating spermatids with spermatozoa, such as in the case of Supplementary Figures 4D3 and D4.

Thanks for the comments, some spermatogenic cells were indeed mislabeled as you mentioned. We have therefore readjusted the labeling accordingly. We also changed spermatozoa to mature spermatids. The new sentence is now: “At the cellular level, fluorescence was detectable in germ cells (B1-B3) including Spermatogonia (Sg), Spermatocytes (Scytes), round Spermatids (RStids), mature spermatids (m-Sptids) and Sertoli cells (SC)”. Moreover, to indicate the localization of the basal membrane, we have also labelled myoid cells.

The characterization of Armc2 expression could have been improved as well. The authors show a convincing expression of ARMC2 in a few spermatids/sperm using a combination of an anti-ARMC2 antibody and tubules derived from ARMC2 KO animals. At the minimum, one would have liked to see at least one whole tubule of a relevant stage.

Thanks for the remark.

We present now new images showing transversal section of seminiferous tubules as requested (see supp fig 6). In this new figure, it is clear that Armc2 is only expressed in spermatids. We have also added in this figure an analysis of the RNA-seq database produced by Gan's team (Gan, Wen et al. 2013), confirming that ArmC2 expression is predominantly expressed at the elongated spermatid stage. This point is now clearly indicated in the text.

Overall, the authors show that electroporating mRNA can improve spermatogenesis as demonstrated by the generation of motile sperm in the ARMC2 KO mouse model.

Thank you

Reviewer #2 (Public Review):

Summary:

Here, the authors inject naked mRNAs and plasmids into the rete testes of mice to express exogenous proteins - GFP and later ARMC2. This approach has been taken before, as noted in the Discussion to rescue Dmc1 KO infertility. While the concept is exciting, multiple concerns reduce reviewer enthusiasm.

Strengths:

The approach, while not necessarily novel, is timely and interesting. Weaknesses:

Overall, the writing and text can be improved and standardized - as an example, in some places in vivo is italicized, in others it's not; gene names are italicized in some places, others not; some places have spaces between a number and the units, others not. This lack of attention to detail in the preparation of the manuscript is a significant concern to this reviewer - the presentation of the experimental details does cast some reasonable concern with how the experiments might have been done. While this may be unfair, it is all the reviewers have to judge. Multiple typographical and grammatical errors are present, and vague or misleading statements.

Thanks for the comment, we have revised the whole manuscript to remove all the mistakes. We have also added new experiments/figures to strengthen the message. Finally, we have substantially modified the discussion.

Reviewer #3 (Public Review):

Summary:

The authors used a novel technique to treat male infertility. In a proof-of-concept study, the authors were able to rescue the phenotype of a knockout mouse model with immotile sperm using this technique. This could also be a promising treatment option for infertile men.

Strengths:

In their proof-of-concept study, the authors were able to show that the novel technique rescues the infertility phenotype in vivo.

Weaknesses:

Some minor weaknesses, especially in the discussion section, could be addressed to further improve the quality of the manuscript.

We have substantially modified the discussion, following the remarks of the reviewers.

It is very convincing that the phenotype of Armc2 KO mice could (at least in part) be rescued by injection of Armc2 RNA. However, a central question remains about which testicular cell types have been targeted by the constructs. From the pictures presented in Figures 7 and 8, this issue is hard to assess. Given the more punctate staining of the DNA construct a targeting of Sertoli cells is more likely, whereas the more broader staining of seminiferous tubules using RNA constructs is talking toward germ cells. Further, the staining for up to 119 days (Figure 5) would point toward an integration of the DNA construct into the genome of early germ cells such as spermatogonia and/or possibly to Sertoli cells.

Thanks for the comment. We would like to recall the peculiar properties of the non-insertional Enhanced Episomes Vector (EEV) plasmid, which is a non-viral episome based on the Epstein-Barr virus (EBV: Epstein-Barr Virus). It allows the persistence of the plasmid for long period of time without integration. Its maintenance within the cell is made possible by its ability to replicate in a synchronous manner with the host genome and to segregate into daughter cells. This is due to the fact that EEV is composed of two distinct elements derived from EBV: an origin of replication (oriP) and an EpsteinBarr Nuclear Antigen 1 (EBNA1) expression cassette (Gil, Gallaher, and Berk, 2010). The oriP is a locus comprising two EBNA1-binding domains, designated as the Family of Repeats (FR) and Dyad Symmetry (DS). The FR is an array of approximately 20 EBNA1-binding sites (20 repeats of 30 bp) with high affinity, while the DS comprises four lower-affinity sites operating in tandem (Ehrhardt et al., 2008).

The 641-amino-acid EBNA1 protein contains numerous domains. The N-terminal domains are rich in glycines and alanines, which enable interaction with host chromosomes. The C-terminal region is responsible for binding to oriP (Hodin, Najrana, and Yates, 2013). The binding of EBNA1 to the DS element results in the recruitment of the origin of replication. This results in the synchronous initiation of extra-chromosomal EEV replication with host DNA at each S phase of the cell cycle (Düzgüneş, Cheung, and Konopka 2018). Furthermore, EBNA1 binding to the FR domain induces the formation of a bridge between metaphase chromosomes and the vector during mitosis. This binding is responsible for the segregation of the EEV episome in daughter cells (Düzgüneş, Cheung, and Konopka 2018). It is notable that EEV is maintained at a rate of 90-95% per cell division.

Because of the intrinsic properties of EEV described above, the presence of the reporter protein at 119 day after injection was likely due to the maintenance of the plasmid, mostly in Sertoli cells, and not to the DNA integration of the plasmid.

Of note, the specificity of EEV was already indicated in the introduction (lines 124-128 clean copy). Nevertheless, we have added more information about EEV to help the readers.

Given the expression after RNA transfection for up to 21 days (Figure 4) and the detection of motile sperm after 21 days (Figure 11), this would point to either round spermatids or spermatocytes. These aspects need to be discussed more carefully (discussion section: lines 549-574).

We added a sentence to highlight that spermatids are transfected and protein synthesized at this stage and this question is discussed in details (see lines 677-684 clean copy).

It would also be very interesting to know in which testicular cell type Armc2 is endogenously expressed (lines 575-591)

Thanks for the remarks. We present now new images showing the full seminiferous tubules as requested by reviewer 1 (see supp fig 6). In this new figure, it is clear that Armc2 is only expressed in spermatids. We have also added in this figure an analysis of the RNA-seq database produced by Gan's team (Gan, Wen et al. 2013), confirming that Armc2 is predominantly expressed at the elongated spermatid stage. This point is now clearly indicated in the text. (lines 570-579 clean copy).

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

The article is well-structured and easy to read. Nonetheless, there are typos and mistakes in some places that are distracting to the reader, such as the capitalization of the word "Oligo-" in the title of the manuscript, the use of the word "Materiel" in the title of the Materials and methods and the presence of space holders "Schorr staining was obtained from Merck (XXX)". Thank you, we corrected the misspelling of "Materials and Methods" and corrected our error: "obtained from Merck (Darmstadt, Germany)". We also carefully corrected the manuscript to remove typos and mistakes.

The discussion is too lengthy, with much repetition regarding the methods used and the results obtained. For example, these are two sentences from the discussion. "The vector was injected via the rete testis into the adult Armc2 KO mice. The testes were then electroporated." I would recommend shortening these passages.

Thanks for your comments, we removed the sentences and we have substantially modified the discussion, following the remarks of the reviewers.

The work is extensive, and many experiments have been done to prove the points made. However, a more in-depth analysis of critical experiments would have benefited the manuscript significantly. A more thorough analysis of sperm mobility and morphology using the CASA system would have been an initial step.

In response to the observations made, additional CASA experiments and sperm motility analysis were conducted, as illustrated in Figure 11 (A2-A3). Individual CASA parameters for motile sperm cells were extracted as suggested and represented in a new graph (Fig 11 A2). We have observed significant differences between WT and rescued sperm. In particular, the VSL and LIN parameters were lower for rescued sperm. Nevertheless, these differences were not sufficient to prevent IVF, maybe because the curvilinear velocity (VCL) was not modified.

In the case of ARMC2 localization, an analysis of the different stages of spermatogenesis to show when ARMC2 starts to be expressed.

Thanks for the remarks. This is an important remark pointed out by all reviewers. As explained above, we have performed more experiments. We present now new images showing transversal section of seminiferous tubules as requested (see supp fig 6). In this new figure, it is clear that Armc2 is only expressed in spermatid layers. We have also added in this figure an analysis of the RNA-seq database produced by Gan's team (Gan, Wen et al. 2013), confirming that ArmC2 expression is predominantly expressed at the elongated spermatid stage. This point is now clearly indicated in the text. (lines 575579 clean copy).

Finally, exploring additional endpoints to understand the quality of the sperm generated, such as the efficiency of ICSI or sperm damage, could have helped understand the degree of the recovery.

This point was underlined in public review. We paste here our answer: "To address this important point, the ability of sperm to produce embryos was therefore challenged by two different assisted reproduction technologies, that are IVF and ICSI. To increase the number of motile sperm for IVF experiments, we have injected both testes from one male. We also conducted intracytoplasmic sperm injection (ICSI) experiments, using only rescued sperm, identified as motile sperm with a normal flagellum. The results of these new experiments have demonstrated that the rescued ARMC2 sperm successfully fertilized eggs and produced embryos at the two-cell stage by IVF and blastocysts by ICSI. These outcomes are presented in Figure 12."

Reviewer #2 (Recommendations For The Authors):

38,74 intracellular

Thanks, we changed it accordingly: "Intracytoplasmic sperm injection (ICSI) is required to treat such a condition, but it has limited efficacy and has been associated with a small increase in birth defects" and "such as intracytoplasmic sperm injection (ICSI)".

39 "limited efficacy" Versus what? And for what reason? "small increase in birth defects" - compared to what?

We changed to "... but it is associated with a small increase in birth defect with comparison to pregnancies not involving assisted conception."

40 Just thinking through the logic of the argument thus far - the authors lay out that there are people with OAT (true), ICSI must be used (true), ICSI is bad (not convincing), and therefore a new strategy is needed... so is this an alternative to ICSI? And this is to restore fertility, not "restore spermatogenesis"

- because ICSI doesn't restore spermatogenesis. This logic flow needs to be cleaned up some

Thanks we changed it accordingly: "restore fertility."

45 "mostly"?

Thank you, we removed the word: "We show that mRNA-coded reporter proteins are detected for up to 3 weeks in germ cells, making the use of mRNA possible to treat infertility."

65 Reference missing.

We added the following reference Kumar, N. and A. K. Singh (2015). "Trends of male factor infertility, an important cause of infertility: A review of literature." J Hum Reprod Sci 8(4): 191-196.

68 Would argue meiosis is not a reduction of the number of chromosomes - that happens at the ends of meiosis I and II - but the bulk of meiosis is doubling DNA and recombination; would re-word; replace "differentiation" with morphogenesis, which is much more commonly used:

Thank you, we have changed the sentence accordingly: "proliferation (mitosis of spermatogonia), reduction of the number of chromosomes (meiosis of spermatocytes), and morphogenesis of sperm (spermiogenesis)".

70 "almost exclusively" is an odd term, and a bit of an oxymoron - if not exclusively, then where else are they expressed? Can you provide some sense of scale rather than using vague words like "large", "almost", "several", "strongly" and "most...likely" - need some support for these claims by being more specific:

Thanks for the comment, we changed the sentence: "The whole process involves around two thousand genes, 60% of which are expressed exclusively in the testes."

73 "severe infertility" is redundant - if they are infertile, is there really any more or less about it? I think what is meant is patients with immotile sperm can be helped by ICSI - so just be more specific...

We changed the transition : "Among infertility disorders, oligo-astheno-teratozoospermia (OAT) is the most frequent (50 % (Thonneau, Marchand et al. 1991); it is likely to be of genetic origin. Spermatocytograms of OAT patients show a decrease in sperm concentration, multiple morphological defects and defective motility. Because of these combined defects, patients are infertile and can only conceive by IntraCytoplasmic Sperm Injection (ICSI). IntraCytoplasmic Sperm Injection (ICSI) can efficiently overcome the problems faced. However, there are ..."

75 "some" is vague - how many concerns, and who has them? Be specific!

Thanks for the comment, we removed the word.

76-7 Again, be specific - "real" has little meaning - what is the increased risk, in % or fold? This is likely a controversial point, so make sure you absolutely support your contention with data .

77 "these"? There was only one concern listed - increased birth defects; and "a number" is vague - what number, 1 or 1,000,000? A few (2-3), dozens, hundreds?

Thanks for the comment, we have reworded the sentence: "Nevertheless, concerns persist regarding the potential risks associated with this technique, including blastogenesis defect, cardiovascular defect, gastrointestinal defect, musculoskeletal defect, orofacial defect, leukemia, central nervous system tumors, and solid tumors. Statistical analyses of birth records have demonstrated an elevated risk of birth defects, with a 30–40% increased likelihood in cases involving ICSI, and a prevalence of birth defects between 1% and 4%." We have added a list of references to support these claims.

79-81 So, basically transgenesis? Again, vague terms "widely" - I don't think it's all that widely used yet... and references are missing to support the statement that integration of

DNA into patient genomes is widely used. Give specific numbers, and provide a reference to support the contention.

Thanks for the comment, we removed the word widely and add references.

81-5 Just finished talking about humans, but now it appears the authors have switched to talking about mice - got to let the readers know that! Unless you're talking about the Chinese group that deleted CCR5 in making transgenic humans?

Your feedback is greatly appreciated. In response to your comments, the sentence in question has been amended to provide a more comprehensive understanding. Indeed, the text refers to experiences carried in mice. The revised wording is as follows: “Given the genetic basis of male infertility, the first strategy, tested in mice, was to overcome spermatogenic failure associated with monogenic diseases by delivery of an intact gene to deficient germ cells (Usmani, Ganguli et al. 2013).

84-5 "efficiently" and "high" - provide context so the reader can understand what is meant - do the authors mean the experiments work efficiently, or that a high percentage of cells are transfected? And give some numbers or range of numbers - you're asking the readers to take your word for things when you choose adjectives - instead, provide values and let the readers decide for themselves.

Thanks for the comment, we have reworded the sentence: Gene therapy is effective in germ cells, as numerous publications have shown that conventional plasmids can be transferred into spermatogonia in several species with success, allowing their transcription in all cells of the germinal lineage (Usmani, Ganguli et al. 2013, Michaelis, Sobczak et al. 2014, Raina, Kumar et al. 2015, Wang, Liu et al. 2022).

93 Reference at the end of the sentence "most countries"

Thanks, we changed the sentence and added the reference: the new sentence is "... to avoid any eugenic deviations, transmissible changes in humans are illegal in 39 countries (Liu 2020)" (Liu, S. (2020). "Legal reflections on the case of genomeedited babies." Glob Health Res Policy 5: 24

93-4 Odd to say "multiple" and then list only one.

Thanks for the comment, we have reworded the sentence: “Furthermore, the genetic modification of germ cell lines poses biological risks, including the induction of cancer, off-target effects, and cell mosaicism. Errors in editing may have adverse effects on future generations. It is exceedingly challenging to anticipate the consequences of genetic mosaicism, for instance, in a single individual. (Sadelain, Papapetrou et al. 2011, Ishii 2017).”

97 Is this really a "small" change? Again, would use adjectives carefully - to this reviewer, this is not a small change, but a significant one! And "should be" is not altogether convincing

Thanks for the comment, we have reworded the sentence: “Thanks to this change, the risk of genomic insertion is avoided, and thus there is no question of heritable alterations.”

What chance is there of retrotransposition? Is there any data in the literature for that, after injecting millions of copies of RNA one or more might be reverse transcribed and inserted into the genome?

This is certainly possible and is the putative origin for multiple intronless spermatid-expressed genes:

The expert poses an interesting question, but one that unfortunately remains unanswered at present. Most papers on mRNA therapy state that there is no risk concerning genomic integration, but no reference is given (for instance see mRNA-based therapeutics: looking beyond COVID-19 vaccines. Lancet. 2024 doi: 10.1016/S0140-6736(23)02444-3). This is an important question, which deserves to be evaluated, but is beyond the scope of this manuscript. Nevertheless is remaining very debating (Igyarto and Qin 2024).

98 Odd to say "should be no risk" and then conclude with "there is no question" - so start the sentence with 'hedging', and then end with certainty - got to pick one or the other.

Thanks for the comment, we have reworded the sentence

99 "Complete" - probably not, would delete:

We removed the word: “The first part of this study presents a characterization of the protein expression patterns obtained following transfection of naked mRNA coding for reporter genes into the testes of mice”

101-2 Reference missing, as are numbers - what % of cases?

Thank you, we changed the sentence and added the reference: “Among infertility disorders, oligoastheno-teratozoospermia (OAT) is the most frequent (50 % (Thonneau, Marchand et al. 1991)” Thonneau, P., S. Marchand, A. Tallec, M. L. Ferial, B. Ducot, J. Lansac, P. Lopes, J. M. Tabaste and A. Spira (1991). "Incidence and main causes of infertility in a resident population (1,850,000) of three French regions (1988-1989)." Hum Reprod 6(6): 811-816.

103 Once again, the reference is missing:

We have added these references: (Colpi, Francavilla et al. 2018) (Cavallini 2006)

104-5 Awkward transition.

Thanks, we changed the transition: “The first part of this study presents a characterization of the protein expression patterns obtained following transfection of naked mRNA coding for reporter genes into the testes of mice. The second part is to apply the protocol to a preclinical mouse model of OAT.”

105 Backslash is odd - never seen it used in that way before

Removed

108 "completely infertile" is redundant;

Thank you, we changed it accordingly: “Patients and mice carrying mutations in the ARMC2 gene present a canonical OAT phenotype and are infertile”.

and is a KO mouse really "preclinical"?

The definition of preclinical research, is research involving the use of animals to ascertain the potential efficacy of a drug, procedure, or treatment. Preclinical studies are conducted

prior to any testing in humans. Our KO mouse model has been shown to mimic human infertility. Indeed *Armc2*^{-/-} mice exhibit a phenotype that is identical to that observed in humans. Our study is in line with this definition. For this reason, we have decided to maintain our current position and to use the term "preclinical" in the article.

| 110 Delete "sperm".

Thank you, we changed it accordingly: "The preclinical *Armc2* deficient (*Armc2* KO) mouse model is therefore a valuable model to assess whether in vivo injection of naked mRNA combined with electroporation can restore spermatogenesis"

| 111 "Easy"? Really?

We changed it accordingly: "We chose this model for several reasons: first, *Armc2* KO mice are sterile and all sperm exhibit short, thick or coiled flagella [13]."

| 112-3 "completely immobile" is redundant - either they are immobile or not.

Thank you, we changed it accordingly: "As a result, 100 % of sperm are immobile, thus it should be easy to determine the efficacy of the technique by measuring sperm motility with a CASA system."

| 108-33 Condense this lengthy text into a coherent few sentences to give readers a sense of what you sought to accomplish, broadly how it was done, and what you found. This reads more like a Results section

Thanks for the comment, we shortened the text.

| *Materials and Methods*

The sections appear to have been written by different scientists - the authors should standardize so that similar detail and formatting are used - e.g., in some parts the source is in parentheses with catalog number, in others not, some have city, state, country, others do not... the authors should check eLife mandates for this type of information and provide.

We are grateful for your feedback. We standardized the text, and if we had missed some, as outlined on the E-Life website, we can finish to format the article once it has been accepted for publication in the journal before sending the VOR.

| 134 Misspelling

We corrected the misspelling

| 142 Just reference, don't need to spell it out.

Thanks, we changed it accordingly: "and the *Armc2* KO mouse strain obtained by CRISPR-Cas9 (Coutton, Martinez et al. 2019). Experiments"

| 150 What is XXX?

We would like to express our gratitude for bringing this error to our attention. We have duly rectified the issue: "obtained from Merck (Darmstadt, Germany)."

157-60 Are enough details provided for readers to repeat this if necessary? Doesn't seem so to this reviewer; if kits were followed, then can say "using manufacturer's protocol", or refer to another manuscript - but this is too vague.

Thanks, we change it accordingly: After expansion, plasmids were purified with a NucleoBond Xtra Midi kit (740410-50; Macherey-Nagel, Düren, Germany) using manufacturer's protocol."

165 Again, too few details - how was it purified? What liquid was it in?

Thanks for the comment, the EEV plasmids were purified like all other plasmids. We change the text: "All plasmids,EEV CAGs-GFP-T2A-Luciferase,((EEV604A-2), System Bioscience, Palo Alto, CA, USA), mCherry plasmid (given by Dr. Conti MD at UCSF, San Francisco, CA, USA) and EEV-Armc2-GFP plasmid (CUSTOM-S017188-R2-3,Trilink,San Diego, USA) were amplified by bacterial transformation"

170 Seems some words are missing - and will everyone know Dr. Conti by last name alone? Would spell out, and the details of the plasmid must either be provided or a reference given; how was amplification done? Purification? What was it resuspended in?

Thank for the remark, the mcherry plasmids were purified like all other plasmids. We change the text: "All plasmids,EEV CAGs-GFP-T2A-Luciferase,((EEV604A-2), System Bioscience, Palo Alto, CA, USA), mCherry plasmid (given by Dr. Conti MD, UCSF, San Francisco, CA, USA) and EEV-Armc2-GFP plasmid (CUSTOM-S017188-R2-3,Trilink,San Diego, USA) were amplified by bacterial transformation"

175 Again, for this plasmid provide more information - catalog number, reference, etc; how amplified and purified, what resuspension buffer?

Thank you for the remark, as We mentioned, we add this sentence for the preparation: "All plasmids, EEV CAGs-GFP-T2A-Luciferase,((EEV604A-2), System Bioscience, Palo Alto, CA, USA), mCherry plasmid (given by Dr. Conti MD at UCSF, San Francisco, CA, USA) and EEV-Armc2-GFP plasmid (CUSTOMS017188-R2-3,Trilink,San Diego, USA) were amplified by bacterial transformation" and we add these sentence "The EEV-Armc2-GFP plasmid used for in vivo testes microinjection and electroporation was synthesized and customized by Trilink (CUSTOM-S017188-R2-3,San Diego, USA)."

183 What sequence, or isoform was used? Mouse or human?

Thanks, we changed accordingly: "This non-integrative episome contains the mice cDNA sequences of Armc2 (ENSMUST00000095729.11)"

186-7 Provide sequence or catalog number; what was it resolubilized in?

Thanks we changed accordingly "the final plasmid concentration was adjusted to 9 µg µL-1 in water." We provided the sequence of EEV-Armc2-GFP in supp data 6.

207-219 Much better, this is how the entire section needs to be written!

237-240 Font

Thanks for the comment, we changed it accordingly

| 246 *Cauda, and sperm, not sperm cells*

Thanks for the comment, we changed it accordingly

| 255-6 *Which was done first? Would indicate clearly.*

Thanks for the comment, we changed the sentence: “Adult mice were euthanized by cervical dislocation and then transcardiac perfused with 1X PBS”

| 281-2 *Provide source for software - company, location, etc:*

We changed it accordingly: FIJI software (Opened source software) was used to process and analyze images and Imaris software (Oxford Instruments Tubney Woods, Abingdon, Oxon OX13 5QX, UK) for the 3D reconstructions.

| 323 *um, not uM.*

Thanks for the comment, we changed our mistake: “After filtration (100 µm filter)”

| *Results*

| 369 *Weighed.*

Thanks for the comment, we changed our mistake: “the testes were measured and weighed”

| 371 *No difference in what, specifically?*

Thanks for the comment, we changed the sentence to: “No statistical differences in length and weight were observed between control and treated testes”

| 375 *"was respected"? What does this mean?*

Thanks for the comment, we changed the sentence to “The layered structure of germ cells were identical in all conditions”

| 378 *This is highly unlikely to be true, as even epididymal sperm from WT animals are often defective - the authors are saying there were ZERO morphological defects? Or that there was no difference between control and treated? Only showing 2-3 sperm for control vs treatment is not sufficient.*

Your observation that the epididymal spermatozoa from wild-type animals exhibited defective morphology is indeed true. The prevalence of these defects varies by strain, with an average incidence of 20% to 40% (Kawai, Hata et al., 2006; Fan, Liu et al., 2015). To provide a more comprehensive representation, we conducted a Harris-Shorr staining procedure and included a histogram of the percentage of normal sperm in each condition (new figure 2F4). Furthermore, Harris-Shorr staining of the epididymal sperm cells revealed that there were no discernible increases in morphological defects when mRNA and EEV were utilized, in comparison with the control. We add the sentence “At last, Harris-Shorr staining of the epididymal sperm cells demonstrated that there were no increases in morphological defects when mRNA and EEV were used in comparison with the control”.

| 379 *"safe" is not the right word - better to say "did not perturb spermatogenesis".*

Thanks, we changed it accordingly: “these results suggest that in vivo microinjection and electroporation of EEV or mRNA did not perturb spermatogenesis”

| 382-3 *This sentence needs attention, doesn't make sense as written:*

Thanks for the remark, we changed the sentence to: “No testicular lesions were observed on the testes at any post injection time”

| 389 *How long after injection?*

Thanks for the comment, we changed the sentence to: “It is worth noting that both vectors induced GFP expression at one day post-injection”

| 390 *Given the duration of mouse spermatogenesis (~35 days), for GFP to persist past that time suggests that it was maintained in SSCs? How can the authors explain how such a strong signal was maintained after such a long period of time? How stable are the episomally-maintained plasmids, are they maintained 100% for months? And if they are inherited by progeny of SSCs, shouldn't they be successively diluted over time? And if they are inherited by daughter cells such that they would still be expressed 49 days after injection, shouldn't all the cells originating from that SSC also be positive, instead of what appear to be small subsets as shown in Fig. 3H2? Overall, this reviewer is struggling to understand how a plasmid would be inherited and passed through spermatogenesis in the manner seen in these results.*

Thanks for the comment.

This point was already underlined in public review. We paste here our answer: “The non-insertional Enhanced Episomes Vector (EEV) plasmid is a non-viral episome based on the Epstein-Barr virus (EBV: Epstein-Barr Virus). Its maintenance within the cell is made possible by its ability to replicate in a synchronous manner with the host genome and to segregate into daughter cells. This is due to the fact that EEV is composed of two distinct elements derived from EBV: an origin of replication (oriP) and an Epstein-Barr Nuclear Antigen 1 (EBNA1) expression cassette (Gil, Gallaher, and Berk, 2010). The oriP is a locus comprising two EBNA1-binding domains, designated as the Family of Repeats (FR) and Dyad Symmetry (DS). The FR is an array of approximately 20 EBNA1-binding sites (20 repeats of 30 bp) with high affinity, while the DS comprises four lower-affinity sites operating in tandem (Ehrhardt et al., 2008).

The 641-amino-acid EBNA1 protein contains numerous domains. The N-terminal domains are rich in glycines and alanines, which enable interaction with host chromosomes. The C-terminal region is responsible for binding to oriP (Hodin, Najrana, and Yates, 2013a). The binding of EBNA1 to the DS element results in the recruitment of the origin of replication. This results in the synchronous initiation of extra-chromosomal EEV replication with host DNA at each S phase of the cell cycle (Düzgüneş, Cheung, and Konopka 2018a). Furthermore, EBNA1 binding to the FR domain induces the formation of a bridge between metaphase chromosomes and the vector during mitosis. This binding is responsible for the segregation of the EEV episome in daughter cells (Düzgüneş, Cheung, and Konopka 2018b). It is notable that EEV is maintained at a rate of 90-95% per cell division.”

Because of the intrinsic properties of EEV described above, the presence of the reporter protein at 119 day after injection was likely due to the maintenance of the plasmid, mostly in Sertoli cells, and not to the DNA integration of the plasmid.

Of note, the specificity of EEV was already indicated in the introduction. Nevertheless, we have added more information about it to help the readers (lines 124-128 clean copy)

398 Which "cell types"?

Your feedback is greatly appreciated, and the sentence in question has been amended to provide a more comprehensive understanding. The revised wording is as follows: These results suggest that GFPmRNA and EEV-GFP targeted different seminiferous cell types, such as Sertoli cells and all germline cells, or that there were differences in terms of transfection efficiency.

409 Why is it important to inject similar copies of EEV and mRNA? Wouldn't the EEV be expected to generate many, many more copies of RNA per molecule than the mRNAs when injected directly??

We removed the word importantly.

415 How is an injected naked mRNA stably maintained for 3 weeks? What is the stability of this mRNA?? Wouldn't its residence in germ cells for 21 days make it more stable than even the most stable endogenous mRNAs? Even mRNAs for housekeeping genes such as actin, which are incredibly stable, have half-lives of 9-10 hours.

We appreciate your inquiry and concur with your assessment that mRNA stability is limited. It is our hypothesis that the source of the confusion lies in the fact that we injected mRNA coding for the GFP protein, rather than mRNA tagged with GFP. After a three-week observation period, we did not observe the mRNA, but we observed the expression of the GFP protein induced by the mRNA. To draw the reader's attention to this point, we have added the following sentence to the text "It is important to underline that the signal measured is the fluorescence emitted by the GFP. This signal is dependent of both the half-lives of the plasmid/mRNA and the GFP. Therefore, the kinetic of the signal persistence (which is called here expression) is a combination of the persistence of the vector and the synthesized protein. See lines 469-472 clean copy.

This being said, it is difficult to compare the lifespan of a cellular mRNA with that of a mRNA that has been modified at different levels, including 5'Cap, mRNA body, poly(A)tail modifications, which both increase mRNA stability and translation (see The Pivotal Role of Chemical Modifications in mRNA Therapeutics (2022) <https://doi.org/10.3389/fcell.2022.901510>). This question is discussed lines 687-698 clean copy

467 "safely" should be deleted

Thanks, we removed the word: "To validate and confirm the capacity of naked mRNA to express proteins in the testes after injection and electroporation"

470 Except that apoptotic cells were clearly seen in Figure 2:

We would like to thank the reviewer for their comment. We agree that the staining of the provided sections were of heterogenous quality. To address the remark, we carried out additional HE staining for all conditions, and we now present testis sections correctly stained obtained in the different condition in Fig. 2 and Supp. 7. Our observations revealed that the number of apoptotic cells remained consistent across all conditions.

471 "remanence"?

We appreciate your feedback and have amended the sentence to provide clear meaning. The revised wording is as follows: "The assessment of the temporal persistence of testicular

mCherry fluorescent protein expression revealed a robust red fluorescence from day 1 post-injection, which remained detectable for at least 15 days (Fig. Supp. 3 B2, C2, and D2)."

489 IF measures steady-state protein levels, not translation; should say you determined when ARMC2 was detectable.

Thanks for the remark, we changed the sentence to: " By IF, we determined when ARMC2 protein was detectable during spermatogenesis."

491 Flagella

Thanks for the comment, we changed our mistake: "in the flagella of the elongated spermatids (Fig 9A)"

Discussion

The Discussion is largely a re-hashing of the Methods and Results, with additional background.

Message stability must be addressed - how is a naked mRNA maintained for 21 days?

As previously stated, it is our hypothesis that the source of the confusion lies in the fact that we injected mRNA coding for the GFP protein, rather than mRNA tagged with GFP. After a three-week observation period, we did not observe the mRNA, but we observed the synthesized GFP protein. This point and the stability of protein in the testis is now discussed lines 677-684 (clean copy).

556 How do the authors define "safe"?

Thanks for the comment, we changed the sentence to be clearer: "Our results also showed that the combination of injection and electroporation did not perturb spermatogenesis when electric pulses are carefully controlled"

563 Synthesized

Thanks, we changed it accordingly

602 Again, this was not apparent, as there were more apoptotic cells in Fig. 2 - data must be provided to show "no effect".

As previously stated, we carried out additional HE staining for all conditions, as can be observed in Fig. 2 . Our observations revealed that the number of apoptotic cells remained consistent across all conditions.

629-30 This directly contradicts the authors' contention in the Introduction that ICSI was unsafe - how is this procedure going to be an advancement over ICSI as proposed, if ICSI needs to be used?? Why not just skip all this and do ICSI then?? Perhaps if this technique was used to 'repair' defects in spermatogonia or spermatocytes, then that makes more sense. But if ICSI is required, then this is not an advancement when trying to rescue a sperm morphology/motility defect.

In light of the latest findings (Fig 12), we have revised this part of the discussion and this paragraph no longer exist.

Nevertheless, to address specifically the reviewer's remark, we would like to underline that ICSI with sperm from fertile donor is always more efficient than ICSI with sperm from patient suffering of OAT condition. Our strategy, by improving sperm quality, will improve the efficiency of ICSI and at the end will increase the live birth rate resulting from the first fresh IVF cycle.

640-2 What is meant by "sperm organelles" And what examples are provided for sperm proteins being required at or after fertilization?

This paragraph was also strongly modified and the notion of protein persistence during spermatogenesis was discussed in the paragraph on fluorescent signal duration. See lines 698-705.

651 "Dong team"??

Thanks for the comment, we added the references.

Figure 2D2 - tubule treated with EEV-GFP appears to have considerably more apoptotic cells - this reviewer counted ~10 vs 0 in control; also, many of the spermatocytes appear abnormal in terms of their chromatin morphology - the authors must address this by staining for markers of apoptosis - not fair to conclude there was no difference when there's a very obvious difference!

We would like to thank the reviewer for their comment. This point was already addressed. As previously stated, we provide now new testis sections for all condition (see Fig. 2). Our observations revealed that the number of apoptotic cells remained consistent across all conditions.

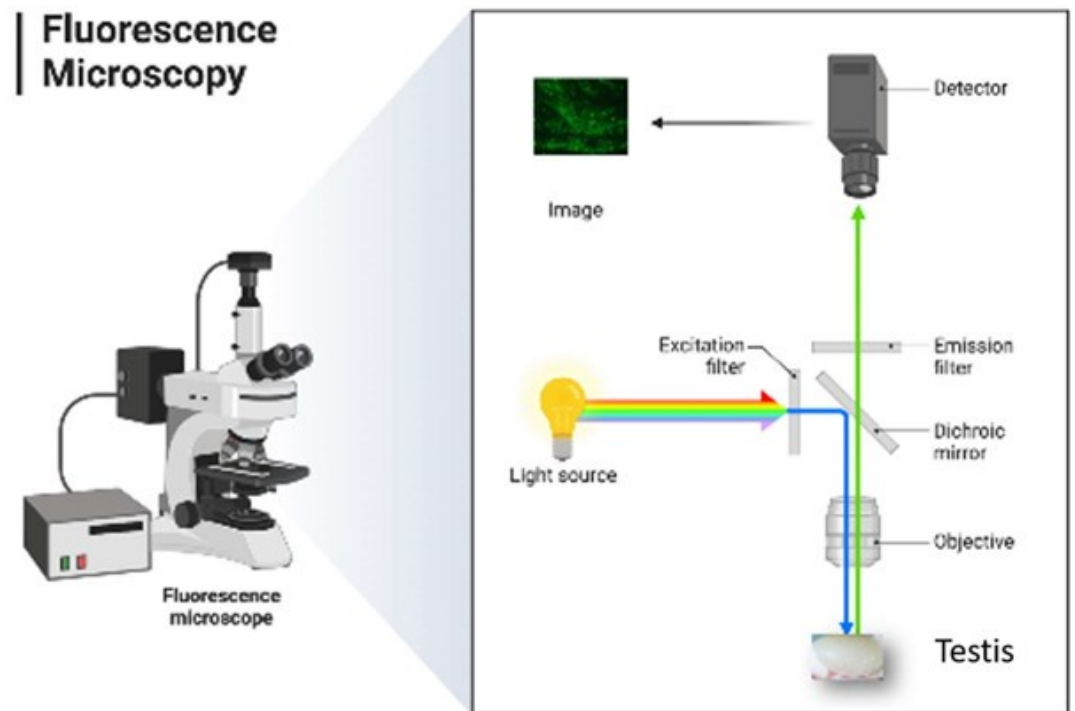
Figure 2D3 staining is quite different than D1-2, likely a technical issue - looks like no hematoxylin was added? Need to re-stain so results can be compared to the other 2 figures

As previously stated, we carried out additional HE staining for all conditions, and new images are provided, with similar staining.

Figure 3 - the fluorescent images lack any context of tubule structure so it is nearly impossible to get a sense of what cells express GFP, or whether they're in the basal vs adluminal compartment - can the authors outline them? Indicate where the BM and lumen are.

We would like to thank the reviewer for their comment. This figure provides actually a global view of the green fluorescent protein (GFP) expression at the surface of the testis. The entire testis was placed under an inverted epifluorescence microscope, and a picture of the GFP signal was recorded. For this reason, it is impossible to delineate the BM and the lumen. It should be noted that the fluorescence likely originates from different seminiferous tubules.

Author response image 1.



So, for Figure 3 if the plasmid is being uptaken by cells and maintained as an episome, is it able to replicate? Likely not.

Yes! it is the intrinsic property of the episome, see the detailed explanation provided above about the EEV plasmid

So, initially, it could be in spermatogonia, spermatocytes, and spermatids. As time progressed those initially positive spermatids and then spermatocytes would be lost - and finally, the only cells that should be positive would be the progeny of spermatogonia that were positive - but, as they proliferate shouldn't the GFP signal decline?

Because EEV is able to replicate in a synchronous manner with the host genome and to segregate into daughter cells at a level of 90% of the mother cell, the expected decline is very slow.

And, since clones of germ cells are connected throughout their development, shouldn't the GFP diffuse through the intercellular bridges so entire clones are positive? Was this observed?

We did not perform IF experiments further than 7 days after injection, a time too short to observe what the reviewer suggested. Moreover, if at 1 day after injection, GFP synthesized from injected EEV was found in both germ cells and Sertoli cells (Fig 7), after one week, the reporter proteins were only observable in Sertoli cells. This result suggests that EEV is maintained only in Sertoli cells, thus preventing the observation of stained clones.

Can these sections be stained for the ICB TEX14 so that clonality can be distinguished? Based on the apparent distance between cells, it appears some are clones, but many are

not...

We thank the reviewer for this suggestion but we are not able to perform testis sectioning and costaining experiments because the PFA treatment bleaches the GFP signal. We also tested several GFP antibodies, but all failed.

Nevertheless, we were able to localize and identify transfected cells thank to the whole testis optical clearing, combined with a measure of GFP fluorescence and three-dimensional image reconstructions.

For Figure 4, with the mRNA-GFP, why does the 1-day image (which looks similar to the plasmidtransfected) look so different from days 7-21?

And why do days 7-21 look so different from those days in Fig 3?

Thank you for your feedback. It is an excellent question. Because of the low resolution of the whole testis epifluorescences imaging and light penetration issue, we decided to carry-out whole testis optical clearing and three-dimensional image reconstructions experiments, in order to get insights on the transfection process. At day 1, GFP synthesized from EEV injection was found in spermatogonia, spermatocytes and Sertoli cells (Fig 7). After one week, the reporter protein synthesized from injected EEV was only observable in Sertoli cells.

In contrast, for mRNA, on day 1 and day 7 post-injection, GFP fluorescent signal was associated with both Sertoli cells and germ cells. This explains why patterns between mRNA-GFP and EEV-GFP are similar at day 1 and different at day 7 between both conditions.

Why do the authors think the signal went from so strong at 21 to undetectable at 28? What changed so drastically over those 7 days?

What is the half-life of this mRNA supposed to be? It seems that 21 days is an unreasonably long time, but then to go to zero at 28 seems also odd... Please provide some explanation, and context for whether the residence of an exogenous mRNA for 21 days is expected.

As previously stated, it is our hypothesis that the source of the confusion lies in the fact that we injected mRNA coding for the GFP protein, rather than mRNA tagged with GFP. After a three-week observation period, we did not observe the mRNA, but we observed the GFP protein produced by the mRNA. The time of observation of the reporter proteins expressed by the respective mRNA molecules (mCherry, luciferase, or GFP) ranged from 15 to 21 days. Proteins have very different turnover rates, with half-lives ranging from minutes to days. Half-lives depend on proteins but also on tissues. As explained in the discussion, it has been demonstrated that proteins involved in spermatogenesis exhibit a markedly low turnover rate and this explains the duration of the fluorescent signal.

The authors should immunostain testis sections from controls and those with mRNA and plasmid and immunostain with established germ cell protein fate markers to show what specific germ cell types are GFP+

Thank you for your feedback. As previously mentioned, we were unable to perform testis sectioning and co-staining because the PFA treatment bleaches the GFP signal and because we were unable to reveal GFP with an GFP antibody, for unknown reasons.

For the GFP signal to be maintained past 35 days, the plasmid must have integrated into SSCs - and for that to happen, the plasmid would have to cross the blood-testis-barrier... is this expected?

We are grateful for your observation.

First, as explained above, we do not think that the plasmid has been integrated.

Concerning the blood-testing barrier. It bears noting that electroporation is a technique that is widely utilized in biotechnology and medicine for the delivery of drugs and the transfer of genes into living cells (Boussetta, Lebovka et al. 2009). This process entails the application of an electric current, which induces the formation of hydrophilic pores in the lipid bilayer of the plasma membrane (Kanduser, Miklavcic et al. 2009). The pores remain stable throughout the electroporation process and then close again once it is complete. Consequently, as electroporation destabilizes the cell membrane, it can also destabilize the gap junctions responsible of the blood-testis barrier. This was actually confirmed by several studies, which have observed plasmid transfection beyond the blood-testis barrier with injection into rete testis following electroporation (Muramatsu, Shibata et al. 1997, Kubota, Hayashi et al. 2005, Danner, Kirchhoff et al. 2009, Kanduser, Miklavcic et al. 2009, Michaelis, Sobczak et al. 2014).

Figure 9 - authors should show >1 cell - this is insufficient; also, it's stated it's only in the flagella, but it also appears to be in the head as well. And is this just the principal piece?? And are the authors sure those are elongating vs condensing spermatids? Need to show multiple tubules, at different stages, to make these claims

We have partly answered to this question in the public review; We paste here our answer

“We present now new images showing the full seminiferous tubules as requested (see supp fig 6). In this new figure, it is clear that Armc2 is only expressed in spermatids. We have also added in this figure an analysis of the RNA-seq database produced by Gan's team (Gan, Wen et al. 2013), confirming that Armc2 expression is predominantly expressed at the elongated spermatid stage. This point is now clearly indicated in the text.”

Concerning the localization of the protein in the head, we confirm that the base of the manchette is stained but we have no explanation so far. This point is now indicated in the manuscript.

Figure 10B2 image - a better resolution is necessary

We are grateful for your feedback. We concede that the quality of the image was not optimal. Consequently, We have replaced it with an alternative.

Figure 11 - in control, need to show >1 sperm; and lower-mag images should be provided for all samples to show population-wide effects; showing 1 "normal" sperm per group (white arrows) is insufficient:

We are grateful for your feedback. We conducted further experiments and provide now additional images in Supp. figure 8.

Reviewer #3 (Recommendations For The Authors):

In this study, Vilpreux et al. developed a microinjection/electroporation method in order to transfect RNA into testicular cells. The authors studied several parameters of treated testis and compared the injection of DNA versus RNA. Using the injection of Armc2 RNA into mice with an Armc2 knockout the authors were able to (partly) rescue the fertility phenotype.

Minor points.

Figure 6 + lines 553+554: might it be that the staining pattern primarily on one side of the testis is due to the orientation of the scissor electrode during the electroporation procedure and the migration direction of negatively charged RNA molecules (Figure 6)?

Your input is greatly appreciated. We concur that the observed peripheral expression is due to both the electroporation and injection. Accordingly, we have amended the sentence as follows: "The peripheral expression observed was due to the close vicinity of cells to the electrodes, and to a peripheral dispersal of the injected solution, as shown by the distribution of the fluorescent i-particles NIRFiP-180."

Discussion of the safety aspect (lines 601-608): The authors state several times that there are no visible tissue changes after the electroporation procedure. However, in order to claim that this procedure is "safe", it is necessary to examine the offspring born after microinjection/electroporation.

Your input is greatly appreciated. Consequently, the term "safe" has been replaced with "did not perturb spermatogenesis" in accordance with the provided feedback. Your assertion is correct; an examination of the offspring born would be necessary to ascertain the safety of the procedure. Due to the quantity of motile sperm obtained, it was not possible to produce offspring through natural mating. However, novel *Armcd2*^{-/-}-rescued sperm samples have been produced and in vitro fertilization (IVF) and intracytoplasmic sperm injection (ICSI) experiments have been conducted. The results demonstrate that the *Armcd2*^{-/-}-rescued sperm can successfully fertilize eggs and produce two-cell embryos by IVF and blastocysts by ICSI. These outcomes are visually represented in Figure 12. The development of embryos up to the blastocyst stage is a step in the right direction.

The discussion section could be shortened. Lines 632-646 are largely a repetition of the introductory section. In addition, the Dong paper (ref. 25) may be interesting; however, this part could also be shortened (lines 647-676). This reviewer would prefer the authors to focus on the technique (different application sites and applied nucleotides) and proof of concept for (partial) phenotype rescue in the knockout mice.

Your contribution is highly valued. In light of your observations and the latest findings, we have substantially revised the discussion accordingly.

Line 63: oocytes rather than eggs.

We are grateful for your input, but we have decided to retain our current position and to use the term "eggs" rather than "oocytes" in our writing because the definition of an oocyte is a female gametocyte or germ cell involved in reproduction. In other words, oocyte corresponds to a germ cell inside the ovary and after ovulation become an egg.

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